PREDICTING ETHYLENE GAS CONCENTRATION USING MACHINE LEARNING

A PROJECT REPORT SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE AWARD OF THE MINOR DEGREE OF

BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING – ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING
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DECLARATION

We, the undersigned, declare that the Mini Project titled **"Predicting Ethylene Gas Concentration Using Machine Learning"** has been carried out by us in partial fulfilment of the requirements for the award of the MINOR degree of **Bachelor of Technology (B.Tech.)** in **COMPUTER SCIENCE AND ENGINEERING – ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING** of **Jawaharlal Nehru Technological University Hyderabad (JNTUH)**, under **VNRVJIET B.Tech. R22 Academic Regulations**, at **Vallurupalli Nageswara Rao Vignana Jyothi Institute of Engineering & Technology, Hyderabad**.

We further declare that this work is original, has not been submitted previously for any degree or similar title, and complies with the Institute's plagiarism and academic ethics guidelines, including AI-generated content. The contributions of all team members are clearly indicated in the report.

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ABSTRACT

Ethylene gas is an indispensable component in agricultural, industrial, and environmental settings, and thus accurate prediction of its concentration is imperative to avoid any losses or hazards. Conventional gas detection sensors are reactionary in nature, prone to errors, and lack selectivity to predict future concentration. In this study, we have proposed a machine learning model based on data analytics to predict the concentration of ethylene gas from multivariate data collected by sensors. The aim of this project is to develop an economical and reliable system that captures the nonlinear relationship between the sensor readings and gas concentration. The methodology comprises several stages, such as data preprocessing, feature selection, and regression modelling of polynomial order. In Phase-II, the complete system is developed, which involves training and optimizing the polynomial model and validating its performance using K-fold cross-validation. Various statistical performance measures were calculated, such as MSE, RMSE, MAE, and R-score. The experimental results suggest that the polynomial model yields satisfactory prediction accuracy and generalizes well to the test data. The two main goals of creating a robust predictive model and addressing the shortcomings of threshold-based detection algorithms have been achieved successfully. The developed system offers a practical solution for real-world applications and can be further extended for integration with IoT platforms for real-time monitoring and control.

Keywords: Ethylene Gas Prediction, Machine Learning, Polynomial Regression, Sensor Data Analysis, Gas Monitoring Systems

TABLE OF CONTENTS

LIST OF FIGURES

LIST OF TABLES

NOMENCLATURE

List all symbols, abbreviations and notations used in the report.

ML	Machine Learning
MSE	Mean Square Error
RMSE	Root Mean Square Error
MAE	Mean Absolute Error

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CHAPTER 1

INTRODUCTION

1.1. Background of the Problem

Ethylene (CH), a simple hydrocarbon gas, finds importance in various fields including agriculture, industrial processes, and environment monitoring. It is known as a naturally occurring plant hormone that influences several biological processes such as fruit ripening, respiration, senescence, and stress response. Because of these attributes ethylene is broadly used in post-harvest management systems to control and regulate the ripening process of fruits and vegetables during storage and transportation. But the same attribute of ethylene gas becomes an important aspect of estimating agricultural produce shelf life and quality. Ethylene concentration, even by a small amount, affects the rate of fruit ripening thus making it necessary to accurately monitor.

Ethylene can over-ripen and degrade the stored agricultural produce in storage units and warehouses which cause significant economic loss in fruits, distributors and retail markets. Fruits like banana, tomato and apple are sensitive to ethylene concentration, improper control of it could increase their deterioration and lead to less value. Thus, precise measurement and controlling the ethylene concentration is needed for smart agricultural systems.

Industrial applications are other significant fields where ethylene has great importance. It is a vital source of fuel for petrochemical industries and a main building block in producing polyethylene, ethylene oxide, and other chemical materials. Though the industrial applications are important, ethylene gas is highly flammable and is an immediate hazard if concentration gets too high in enclosed environment like storage tank, processing unit and chemical plant.

The present gas detecting systems work on the principle of detecting the gas level and providing a signal (alarm) once it exceeds the threshold limit. However, this approach suffers from some limitations as it is a reactive system. They provide the alerts only after the gas concentration exceeds the threshold limit and they do not forecast and warn before that. Also, the sensitivity and selectivity issues in most gas sensors lead to an ambiguity in detecting gas. Environmental factors like humidity, temperature and sensor drift cause a significant noise.

With the advancement of sensor technology, large volumes of data can be acquired through gas sensors in real time. This opened the way to use data driven approaches like machine learning for gas monitoring. Machine learning algorithms can find out nonlinear relationships, unknown patterns and use them for prediction.

One main challenge in ethylene detection can be overcome by the use of multi-purpose gas sensors. Metal oxide semiconductor sensors can detect many types of gases. When they are put together, they exhibit a unique response for each type of gas, these response patterns are analyzed and identified by machine learning algorithms. This method is called as sensor fusion. This method avoids using separate dedicated sensor for each gas type making the solution cost-effective.

A smart gas prediction system requires a properly pre-processed data, extracted features and advanced machine learning models. Preprocessing involves removing noise and abnormalities in the data like outlier, normalization and smoothing. Since nonlinear relationships exist between sensor response and ethylene concentration, polynomial regression can be utilized for prediction. The model also needs to be trained and tested efficiently which can be done through K-Fold cross validation method.

Applications of such smart gas prediction systems are widespread: In agriculture to optimize fruit ripening, in industrial safety for detecting gas leaks, in environment monitoring to estimate air quality etc. IoT enabled smart gas prediction system can provide remote monitoring and control systems.

As seen, need of a robust and predictive gas detection system necessitates the adoption of machine learning approaches. It will overcome the drawbacks of current gas detection systems thus helping to achieve the desired outcomes for agriculture, industry and environment safety and these factors are the background for this project.

1.2 Need and Relevance

Due to the need for effective monitoring in agriculture, industry and environment sectors, the need of accurate and reliable Ethylene gas concentration prediction system has been highly felt in this current age. Ethylene (CH), while beneficial if maintained within specific limits, proves to be of significance when concentration is not in the prescribed range. While it can be a useful plant hormone, it also forms a hazardous gas which must be efficiently monitored in this modern world of technology and industrial sector.

In agriculture, ethylene gas has a very crucial part in ripening fruits and vegetables. While in the controlled condition this gas is used for uniform ripening of fruits and vegetables, if the gas concentration becomes non uniform, the results can be over-ripening and spoilage. In storage places like cold storage, warehouses, transporting units, slight deviations in Ethylene concentration can affect tons of fruits and vegetables. Food demand across the globe has increased, thereby requiring efficient management in reducing post-harvest losses. Hence, there is a high requirement for an intelligent system that can not only predict the concentration of gas but also predict it earlier to manage storage more effectively.

In the industrial sector, it serves as a feedstock in petro-chemical industries for production of plastics, polymers and other industrial products. However, in high concentration levels Ethylene becomes an explosion hazard and also risks workers' health. In plants like chemical, refineries etc, a leakage of the gas can cause dangerous circumstances. Traditional gas detection systems rely upon threshold levels for detection of gas concentration and thus require an efficient predictive system before its dangerous level is reached to effectively prevent accidents and potential disasters in industrial sector.

In environmental monitoring, Ethylene is one of the several Volatile Organic Compounds (VOC) that contributes to air pollution and thus its concentration level in the ambient air is an important factor in air quality monitoring. The increasing concerns over environment and its protection and with widespread use of plants in controlled environment have led to necessity of monitoring it.

The importance of the project has also been highlighted from the lack of reliable methods to monitor the Ethylene concentration. Conventional systems are usually designed based on specific sensors which are costly, requires regular calibration and are not able to adapt to different environments. Most importantly these systems operate on threshold-based detection which only alerts on detection of a particular concentration of gas and lacks the capacity to analyze the future trend of the gas concentration. Machine learning techniques can prove to be a versatile and cheap substitute that can effectively learn from non-specific sensors and can be used to develop a reliable ethylene concentration prediction system.

Another reason why the project is relevant is the evolution of the sensors and data acquisition systems. The modern gas sensors produce bulk amount of data and it is necessary to find some way to intelligently process the data. The complex machine learning techniques can handle such large amount of data effectively and also analyze the relation between various factors leading to the changes in gas concentration and predict future levels more accurately.

Thus, machine learning is highly relevant for a multivariate data analysis required in a gas concentration prediction system.

Integration with IoT platform: As discussed, the prediction system can also be a part of an integrated IoT based system, which would allow remote monitoring, control and alerts in applications like smart agriculture, smart industry etc. These intelligent systems make data-based decisions.

Also, due to the development of smart systems, which utilize latest technology such as Industry 4.0, there is an increasing demand of smart sensors and their integration with intelligent system for monitoring. The machine learning based prediction system would be one of them.

In a nut shell, owing to its essentiality in various applications, the need to build a reliable ethylene gas concentration prediction system for an efficient solution for various issues in agriculture, industry and environment sectors is high, due to limitations of traditional sensor technologies and demands for integrated and intelligent monitoring systems.

1.3 Problem Statement

The ethylene (CH) gas is applied to agriculture, industry processing, and environment monitoring. However, accurate prediction and detection of ethylene concentration is still difficult due to limitation of gas sensing and monitoring techniques. The reason behind is how to balance the positive effect of ethylene (used for ripening fruits, and synthesis of industrial product) and negative effect of ethylene (damage the foods, security risk, environmental pollution).

In agricultural applications, one of the main problems is there is no system to continuously monitor and control the ethylene concentration when transporting or storing the agricultural goods. Since fruits and vegetables are very sensitive to ethylene, a slightly change of concentration can speed up ripening and decrease shelf life, leading to great economic lost. A normal monitoring system is not capable of predicting the concentration in the future and cannot provide adequate means to maintain the appropriate storing environment. Lots of agricultural products have to be wasted every year because of overripe and damaged by its accumulation. So, an advance prediction system is expected to solve this problem.

Compared to agriculture application, the problem is much serious in industry application because of the security requirement. Ethylene is a highly flammable gas, which would lead to dangerous situations such as fire and explosion if it accumulated in some confined space, like chemical plant, storage tank, and pipe lines. A traditional gas detection system only triggers an alarm after its concentration goes above the set value (threshold), which means it can provide limited time to preventive action and could hardly prevent accidents. Another issue is they are incapable of analyzing the trend or gradual changing of gas concentration.

The issue also comes from limitation of gas sensing technique. Most commonly used sensors (Metal Oxide Semiconductor sensors (MOS)) are not very selective, and their output value may respond to multiple gas concentrations and be affected by temperature, humidity, noise and sensor drift etc. Thus, it is difficult to use the sensor readings to estimate ethylene concentration with desired precision.

Moreover, the traditional gas detection system involves expensive and dedicated instruments, which may be difficult to implement on a large scale especially for developing countries, and it always requires expensive maintenance and calibration which will increase the cost. Therefore, the desired solution must use normal sensors on a lower cost while accurate to predict.

Given the availability of sensors and sophisticated computational techniques, machine learning is one feasible technique to overcome these problems. However, training an accurate machine learning model for prediction is not easy as the signals from the sensors is multivariate, noisy and nonlinear and it's critical to choose appropriate model for non-linear mapping from the sensor readings to gas concentration and achieve good prediction accuracy with new data.

Hence, the main problem to solve is: how to design an efficient and accurate ethylene gas prediction system which can overcome the disadvantage of traditional threshold detection method and use the multivariate sensor signals while handling their noise and nonlinear nature.

1.4 Objectives

The central aim of this project is the conception and realization of an efficient, reliable, and scalable system for the prediction of ethylene (CH) gas concentration using machine learning approaches. The motivation for such an undertaking arises from the shortcomings of traditional gas detection systems, which primarily function in a reactive mode and lack the capacity to provide predictive intelligence. By harnessing data-driven methods, the project endeavor is to shift from traditional, threshold-based monitoring to intelligent, predictive frameworks that can parse complex sensor readings and estimate ethylene levels with real-time accuracy.

This project will aim to fulfill the following objectives:

1. To train an accurate ML model to predict the concentration of ethylene (CH) gas using multiple sensors.
2. To replace traditional threshold-based detectors with a predictive approach in order to provide advance notice of gas detection and facilitate better decisions.
3. To apply appropriate data preprocessing techniques, including outlier removal, noise filtering and feature scaling, in order to refine the quality of data.
4. To model non-linear relationships among sensor inputs and the target variable, the concentration of ethylene gas, through application of polynomial regression.
5. To use K-fold cross validation to evaluate and test the trained model in regard to its performance using MSE, RMSE, MAE and R-score as evaluation metrics.
6. To build an inexpensive and extensible detector using widely available general-purpose gas sensors rather than sophisticated, costly units.
7. To build a prototype that could be implemented in real world applications like smart agriculture, industrial safety, and environmental monitoring.

1.5 Scope of the Work

The range of this project is focused on the design, development, and evaluation of a machine learning based system to predict the ethylene (CH) gas concentration based on multivariate sensor data. In this work, the main objective is to design a reliable and accurate predictive model capable of predicting gas concentration by considering the readings of the sensors. The range include all the steps in a machine learning pipeline starting from the data acquisition until the model performance validation including preprocessing, model building and evaluation.

The first range of work consists of working with the sensor-based data that contain the information from many gas sensors. In most of the cases they are Metal Oxide Semiconductor (MOS) sensors that gives a response based on the presence of gas in environment. These sensors are known to lack the selectivity toward a specific gas and response to number of gases hence the data obtained is multivariate and highly complex. Within this project, analysis of these data is undertaken with the view of finding the relation between sensor response and the ethylene concentration. The data is assumed to be in an appropriate format like CSV with each record representing the sensor reading at a specific time instance.

The data preprocessing takes up another segment of the project range. Sensor data normally has noise in them and may contain outlier due to disturbances. Also, the sensors may get drifted over time. Hence the preprocessing consists of eliminating outlier using statistical approach (such as Z-score or IQR) and filtering out noise (using techniques like moving averages or Savitzky-Golay filter). Finally, the data must be scaled such that it is normalized. This ensures that the machine learning model built using this data produces reliable result.

Feature engineering and selection form another part of the project range. Since data has measurements from different sensors, it is necessary to find which input variables are more suitable to be used for the prediction of ethylene gas. Correlation study between different sensors and their corresponding measurement is carried out. Redundancy between sensor variables is checked. It is important to realize that by having too many features, it becomes inefficient to train the model; this is why this part of work includes feature selection. Also, the input variables are converted to polynomial form to facilitate the capture of non-linear relation between sensors measurement and the gas concentration by a machine learning algorithm.

The main part of the project range falls under building a machine learning based model that predicts the ethylene gas concentration using the features computed and validated above. In this work, multi-feature polynomial regression model is proposed. This approach allows modeling non-linear relationships while keeping the model complexity within limits for practical applications. The model is trained using the processed sensor readings and then used to predict the concentration values. Various orders of polynomials are tried out to determine the best fit model.

Validation of the model with unseen data and evaluation of its performance are a crucial part of the project range. For the same, k-fold cross validation method is utilized, which guarantees unbiased evaluation of the model by testing it on multiple test subsets of the original data. The performance of the model is evaluated by means of standard performance metrics such as Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R score which represent the degree of deviation of the predicted values from actual values.

Another part of the project range is visualization of results obtained through the developed machine learning model. Graphical representations of different features correlation heatmaps, distribution plots of variables, plot between predicted vs actual, and the variations of error will be represented. These representations provide more insightful understanding of model performance, identify potential patterns and anomalies, and ascertain how the model actually behave.

The range of work regarding the tools and implementation covers primarily the use of the Python programming language, and its several supporting libraries that include NumPy, Pandas, Scikit-learn, Matplotlib, and Seaborn for numerical computations, data manipulation, machine learning operations, and plotting respectively. The system is designed to be modular for ease of maintenance, modification, and extension for possible future enhancements.

The range of application of the developed system is quite broad that it spans across agriculture to industrial sector to even environmental sensing. In agriculture, the system can be used for detecting levels of ethylene in storage so that to regulate fruit ripening processes and minimize post-harvest losses. In industries, the system can alert at the very early stage of gas leakage thus mitigating hazard and explosion incidents. As environmental sensing application, the system can be used to measure air quality of any environment. Such application can be integrated into the IOT platform to perform all the readings, logging, and alert management.

However, the scope of the current work is restricted to the regression based machine learning model, especially polynomial regression model, and therefore models like deep learning and ensemble models were not explored within this work. Also, the performance of the system heavily depends on the quality and quantity of data that are fed into it. Real time application and hardware level implementation are future extension of this project.

In summary, the scope of this project includes building a machine learning based ethylene gas prediction system from data analysis to the performance validation that would prove useful in applications that range from agriculture to the industries. It offers a cost effective, scalable and accurate solution while there remains a lot of scope to develop it into a real-world application.

1.6 Organization of the Report

Structure of the Report: The report has been structured in various chapters for making the reader understand clearly about the work undertaken to determine the prediction of ethylene (CH) gas concentration using the Machine Learning approach. Each of the chapter is designed in such a way that it is informative to explain about the problem to the conclusion.

Chapter 1: Introduction In the initial chapter of the report, the concept of ethylene gas and its application in agriculture, industry, and environment is introduced. The background of the problem, the necessity of the problem, and the relevance of the research work to existing scenario is explained. Moreover, the problem statement, objectives of the study and the scope of the research work is discussed.

Chapter 2: Literature Review In this chapter, the existing research on gas detection and prediction related to ethylene and similar other gases is reviewed. Existing methods and techniques like sensor-based detection and different machine learning techniques are compared based on their pros and cons and finally identified that there are research gaps which need to be addressed, like lack of predictive capabilities, cost of the sensor etc.

Chapter 3: Methodology In this chapter, the complete process is discussed step-by-step to perform the prediction for ethylene gas concentration. The data set used, its characteristics and structure, various techniques to prepare the data set which involve outlier detection and removal, data scaling and transformation, various feature engineering techniques and selection method to obtain important features like polynomial feature expansion is explained in detail. In addition to that, the construction of predictive model which is multi-feature polynomial regression, implementation tool, and validation techniques used to validate the model are discussed.

Chapter 4: Results and Discussion In this chapter, the predicted results are shown in terms of graphs like predicted vs. Actual data and error graphs to check the performance of the developed model based on various metrics like MSE, RMSE, MAE, and R-score. Various model with different polynomial degrees is evaluated and compared to check which model provides optimal results. The obtained results are further analyzed to describe the strength, weakness, and implications.

Chapter 5: Conclusion and Future Scope In this chapter, a summary of the entire research work done is provided and it is explained how the objectives of the research were fulfilled by the use of Machine Learning based on the predicted ethylene gas concentration. Also, the limitations of the system such as reliability on data set, limitations based on models etc., and future scope like implementation using IoT system, extension to other gases etc are discussed to suggest improvement on the present work.

References and Appendices This section contain references that have been used during the work and appendix which consist of extra charts or figures which are not mentioned in the main text of the report

CHAPTER 2

LITERATURE REVIEW

2.1 Overview of the Existing Work

The detection and prediction of gas concentration is an area of significant interest because it ensures safety, contributes to the environment and productivity of industrial and agricultural systems. Ethylene (CH) has become a target gas in this area because it is a beneficial hormone in plants but also a hazardous industrial gas. Existing studies in this area can be classified into traditional gas detection systems, sensor-based systems and machine learning-based predictive systems.

At the beginning, most gas detection systems rely on traditional detection methods by utilizing specialized gas sensors designed for detecting specific gas. Analytical instruments such as gas chromatography (GC) and infrared (IR) spectroscopy have been used due to their high accuracy and sensitivity. However, such methods require high cost, high precision instrument and are difficult to deploy on real time and scale systems. This methods are mainly used in laboratory environment or extremely controlled industrial environment and cannot satisfy other demand for gas detection.

Since the development of new sensors, especially Metal Oxide Semiconductor (MOS) sensor, gas detection system is more widely applied because MOS sensors are cheap, tiny and can detect various gases. The disadvantage of MOS sensor is its low selectivity. Single sensor can response to more than one gases, and it is hard to calculate concentration of ethylene precisely. Hence, researchers tried to use multiple sensors to form sensor array or e-nose to improve the accuracy of detecting gas. With multiple sensors, each gas will have a different signature pattern, then using machine learning method to analyze the pattern and calculate the gas concentration.

More recently, combining machine learning technique with sensor system becomes a good method to overcome drawbacks of other methods. Machine learning algorithms are very effective at modeling nonlinear relation between sensor reading and gas concentration. Supervised learning algorithm such as linear regression, polynomial regression, support vector regression (SVR), Random Forest, Artificial Neural Network (ANN) are widely applied to the problem of gas prediction. Regression based models can be directly used to estimate continuous variable such as concentration of gases.

Polynomial regression has been utilized in numerous works and achieved considerable results for modeling nonlinear relation of sensor reading and gas concentration, because it can express more complex interactions of variables than linear model, and has interpretable nature for relationship between input and output. However, performance of polynomial regression strongly depends on proper selection of polynomial degree; too high degree leads over fitting issue.

On the other hand, other sophisticated learning algorithms, such as Support Vector Machines (SVM) and Ensemble model like Random Forest are used and generally give better accuracy than simple regression models, especially when dealing with high dimension and noisy data. Artificial Neural Networks (ANN) and Deep learning algorithms are showing promise in detecting very complex pattern, but require enormous amounts of data and computational power; their interpretability issue can also be problematic in some applications.

Besides learning model, it is found that data pre-processing plays an important role for the model performance. Outlier elimination, noise filtering, normalization and feature selection are all commonly used methods. Performance of models could be dramatically improved after data is well processed. Cross validation is also used widely to ensure the model perform well in unknown data.

Still, some challenges are there to be solved. Many models developed are used for gas detection instead of prediction. That means that, it can tell you about the gas level at the present, but not forecast the level at the future. Furthermore, dependence on expensive sensors and the absence of scalable system remain issues. High performance for small data set and real-time performance also need attention.

Regarding to ethylene gas, studies were focused on the agriculture application to monitor fruit ripening and store environment as well as industrial safety application. However, efficient, low cost and accurate predictive model are yet to be achieved to predict ethylene concentration by using multiple sensors.

Overall, traditional sensors have moved towards intelligent data-driven models, which incorporate machine learning into sensor system. Much advancement has been achieved in the domain, but high performance, low cost and scalable predictive system still needs research and development. This project is an endeavor on that aspect and use machine learning model for ethylene gas concentration prediction.

2.2 Comparison of Methods/Approaches

Many different methods and approaches are used in gas detection and concentration prediction which are varied based on accuracy, price, complexity, and expandability in real world situations. Comparing the approaches is necessary to identify strengths and weaknesses and the appropriate methods of CH (ethylene) gas concentration prediction. These methods are categorized into traditional analytical approaches, sensor-based detection systems and machine learning based prediction models.

Some of the oldest approaches are traditional analytical methods such as gas chromatography and infrared spectroscopy, both highly accurate in specifying concentration, but require high cost, training and conditions, so are not practical in large applications for real-time measurements such as security systems.

Sensor-based detection systems are more practical and popular. These can include Metal Oxide Semiconductor (MOS), electrochemical and optical sensors and are available at lower cost and size. They can give real-time output but cannot specify exact concentration and do suffer from interference of different gases, and lack ability in predictive functionality. They mostly use a threshold level to predict a potential hazard when a limit is passed.

To overcome single sensor deficiencies and improve predictive accuracy, multiple sensors are integrated to act as a single detector, i.e. An electronic nose (e-nose). Different sensors react to a unique response from the specific gas, but interpretation is still challenging which makes machine learning approach necessary.

Machine learning approaches are suitable for gas concentration prediction using existing data to predict concentrations. There are many techniques available; starting with the simpler ones; Linear Regression is a simple, efficient algorithm but assumes a linear relation between the variables, and does not work well for complex, non-linear data, which is common for sensors.

Polynomial Regression extends Linear Regression to include higher degree term in order to compensate non-linearity, it provides a higher degree of accuracy. It offers a compromise between accuracy and complexity, and is better suited than Linear Regression. The choice of the degree of the polynomial has to be carefully chosen to avoid overfitting which affects generalization.

Support Vector Regression is an advanced machine learning approach capable of handling non-linear data using a kernel trick. It offers a higher prediction accuracy but is difficult to fine-tune the parameter and is computationally demanding.

Ensemble methods such as Random Forest and Gradient Boosting, are widely used for their high performance. These are ensemble models constructed from a number of decision trees and can be very precise and robust, though more complicated.

Most advanced approaches involve Neural Networks and deep learning models. These models are capable of learning complicated, non-linear relationship in large and high dimensional data, they require a lot of data to train, more processing power and long training duration, though they are also complex and difficult to interpret.

Each approach also has to account for noisy sensor data and multivariate sensors. When paired with appropriate pre-processing such as scaling, outlier rejection, and feature selection, machine learning models achieve higher prediction than traditional approaches. Model validation through K-fold cross-validation should also be performed for all models.

Therefore, as demonstrated, traditional analytical methods and basic sensors cannot compete with machine learning approaches in terms of predicting the exact concentration, which necessitates further investigation of ML techniques available for gas prediction. Machine learning approaches using polynomial regression can strike a balance between accuracy, complexity and interpretability and are most suitable for multivariate sensors with non-linear responses and that are typical of CH prediction.

2.3 Critical Analysis of Literature:

The review of current state of gas detection and prediction systems indicates that both sensing technologies and modeling methods for prediction have evolved rapidly. Nevertheless, certain limitations and gaps still exist even with the progress achieved in the field. Therefore, a critical analysis of the literature is essential in order to point out existing drawbacks and show the justification to develop an advanced system to predict CH gas concentration.

One of the major issues that is highlighted by the literature is that conventional analytical methods like gas chromatography and infrared spectroscopy while highly accurate for detecting and measuring gas concentrations are practically not suitable for real-time and large scale applications. These methods require specialized equipments, precise working conditions in a laboratory, and highly skilled personnel. Thus, they are typically used in research laboratory settings and industrial laboratories only, which forms a gap in real-time and large-scale detection such as agriculture or safety applications where continuous measurements are mandatory.

In this context sensor based systems, especially Metal Oxide Semiconductor (MOS) sensors, are quite common as they are compact and low-cost systems which can perform measurements in real-time. However, an important problem associated with this kind of sensors, as pointed out in the literature, is the lack of selectivity. This means that MOS sensors respond to several different gases, causing a cross-sensitivity which can make the measurement of CH gas concentration inaccurate. In addition to the above, the outputs of the sensors are heavily dependent on external factors like temperature, humidity and drift (sensor aging) that introduce noise and variation into the signal. Many existing studies do mention these difficulties but fail to give a definitive solution.

Another important limitation of most of the gas detection systems studied is their threshold based operation. Conventional gas detection systems are set to raise an alarm only if the gas concentration is above a certain threshold. Such a system while being easy to implement and operate has a serious limitation: it is reactive instead of predictive. Having no prediction capability makes it suitable for emergency safety measures but it can barely be useful to control processes. In similar cases for instance like agricultural crop storage or ripening control where predicting the future concentration can be crucial to maintain the quality, such systems lack effectiveness.

The use of machine learning has partially filled some of these gaps and different models like polynomial, linear and multiple linear regression, Support Vector Regression (SVR), Random Forest and Artificial Neural Network (ANN) has been explored in the literature. While being effective, these models too possess certain drawbacks. Linear and multiple linear regressions are too simple to represent complex and nonlinear relations whereas more complex models such as neural networks and random forests are very good but require many training samples and computations. Polynomial regression has shown some promising results in several applications as a medium between the complexity of advanced models and simplicity of basic ones but its effectiveness is highly dependent on the appropriate choice of the polynomial degree and is not guaranteed on new data due to possible over-fitting which is why careful and correct methods for model validation like cross-validation must be applied as frequently as possible, yet many research works do not do it.

The literature also demonstrates that the majority of the work put forward have neglected to take sufficient steps toward data preprocessing. While some studies discuss it as an important aspect some have even completely ignored its relevance and it is important to say that the absence of good preprocessing techniques like filtering or normalization, especially in dealing with noisy and outlier data which is characteristic of sensor data, results in less reliable and accurate models.

Last but not least the scalability and cost issues can not be neglected. Some of the more accurate systems rely on expensive and complex sensors and models and thus cannot be readily available on the market especially to small industries or to be used by consumers or in developing countries. Even though a recent trend of affordable systems utilizing low-cost sensors and complex algorithms is emerging, the field still lacks a standardized way of providing this balance effectively. A few works in the literature have also focused on real time implementation and Internet of Things (IoT), but majority of studies are based on offline analyses and the real-time execution on embedded systems has been largely ignored and it is critical to consider issues related to system integration, data latency, computational efficiency of models for their successful real-time implementation.

In conclusion it has been demonstrated from a critical review of existing literature that while considerable progress has been made in gas detection and prediction systems using sensors, there are significant shortcomings in many approaches used that do not sufficiently provide a means for reliable detection in real-time and to some extent cost effective solutions. In this perspective, an approach that balances Machine learning with low-cost sensor system and effectively deals with issues like data quality, accuracy of the model and deployability is urgently needed and this is exactly what this project aims to do for CH gas prediction.

2.4 Research Gaps Identified:

The comprehensive review and analysis of existing literature on gas detection and prediction systems reveals a number of research gaps which exist and limit current systems, particularly in the prediction of ethylene (CH) gas concentration and indicate the need for an improved solution that is more accurate, reliable and feasible for real-world applications.

One of the main research gaps highlighted was the lack of predictive capacity within existing gas detection systems. Many are based solely on thresholds which only trigger an alert if the concentration of gas exceeds specific levels. Such systems are practical for basic monitoring but are entirely reactive and lack predictive capability, meaning it is difficult to plan for the future in systems such as industrial safety and agriculture where the future concentration of gas needs to be known. Thus, a predictive system is desirable.

A second research gap discovered was the limitation within current sensor technology, specifically in Metal Oxide Semiconductor (MOS) sensors which, whilst cost-effective and easy to install, respond to multiple types of gas, leading to a difficulty in identifying only ethylene. Furthermore, sensors are sensitive to their environment (temperature, humidity) and their sensitivity can degrade over time, meaning sensor readings can be unreliable. Whilst these issues were acknowledged by researchers they are not adequately addressed and this continues to be a research gap.

A further gap identified within the literature was the lack of emphasis on data pre-processing and enhancement. Many of the models that were found were based solely on the predictive models being highly accurate but failed to address the issue of noisy and messy data from sensors that can lead to the machine learning model performing poorly. Thus, a complete data processing system which filters out noisy data is an area of research required.

Another gap identified was in model selection and its complexity. Whilst simple, linear models were easily trainable, non-linear relationships in sensor data cannot be detected as accurately. More advanced, non-linear models such as Artificial Neural Networks require significant processing power and large datasets and can often over fit on limited data making them unsuitable for general application. Thus, an efficient, highly accurate and suitable prediction model is needed which is not overly complex.

Validation of the model and generalization is also not comprehensively addressed in many papers found. Many studies produced highly accurate training data models but did not confirm the effectiveness on data not yet seen by the model. This meant that in many situations the model was likely to have over-fitted the training data making it completely unsuitable for real world application.

One final major research gap found was in the scalability and cost effectiveness of existing systems. Many were complex and required specialist equipment to implement, limiting them to large, industrial applications rather than use in small-scale farms or developing regions. Systems needed to be efficient in use of resources, cost-effective and generally applicable.

An additional gap that could be recognized was the focus of much research on general gas detection, with few on predicting ethylene specifically and the challenges which that presents, for example in controlling crop ripening, as it would mean a specific system designed to predict ethylene was necessary to accurately measure levels.

A final gap that may have been noted by review of this body of work was the lack of an integrated system that incorporates the full process of gas prediction including the development and implementation of the necessary models, suggesting the final aim was a fully functional system rather than a single model or component.

The identified research gaps were lack of predictive capacity, the use of specific sensor technology limitations and poor resolution in sensor readings, data pre-processing requirements, challenges in model selection, over-fitting and lack of generalization of model, poor scalability and cost-effectiveness and a lack of focus on implementing real-time systems for ethylene detection as opposed to simple detection systems. Addressing these points would lead to a successful, efficient and applicable solution to the prediction of ethylene concentration levels

CHAPTER 3

APPROACH / DESIGN METHODOLOGY

3.1 Overview of Phase-I Work and Outcomes

This first phase, Phase-I, involves understanding of the problem, acquiring and analyzing data and building a strong base for the development of the system for predicting the concentration of ethylene (CH) gas. Phase-I is the initial phase where we analyze whether the usage of machine learning algorithms can help to solve the problem, using the sensor values to predict the values. The results that we obtained in Phase-I helped us to develop a robust strategy for Phase-II.

First and foremost step that we underwent in Phase-I was to understand gas sensing technologies and to find out what are the challenges faced in detecting ethylene gas. We researched on different types of gas sensors and found that MOS (Metal Oxide Semiconductor) sensors are predominantly used for gas detection. But MOS sensors have drawbacks, which is they are quite susceptible to the surrounding temperature, humidity, and there is cross sensitivity, although they are readily available and not that costly. So we can not depend solely on raw values provided by MOS sensors without machine learning, to detect the concentration of ethylene gas precisely.

Another crucial part carried out in Phase-I was dataset identification and acquisition. We acquire a multivariate gas sensor dataset that consist of multiple sensor values with gas concentration. We analyzed the dataset with detailed examination to find out the features of the dataset and to tackle with various difficulties in the dataset such as absence of data, noisy data, etc. Exploratory data analysis (EDA) was performed on the dataset using statistical approaches and visualizations. We looked at correlations between sensor values, gas concentration and the relationships in the features.

The data processing was one important part of Phase-I. The sensor values obtained from the gas sensors may be noisy, thus we performed a number of processes on the data so that it may not affect the machine learning algorithms. The processes included finding out and handling the missing values present in the dataset, removing the outliers and noise from the data. Some filtering techniques like smoothing were used to remove the noise from the data. Also, data scaling such as normalization and standardization was applied to the input data.

Feature selection and analysis were also done. We found the correlations between features and the gas concentration. Using correlation, we decided on the important features that are useful for detecting ethylene gas concentration and discard those features that are redundant.

We started with basic model development using linear regression to find a baseline performance on the dataset and found that the relation is not linear. Then we used polynomial regression, to find a non-linear relationship. In fact the performance showed that polynomial regression fits the data well. The insights gained from these preliminary tests helped to determine the most suitable ML approaches for the given task.

We also found suitable evaluation metrics. Evaluation metrics are crucial for performance analysis. Hence we selected Mean squared error (MSE), root mean squared error (RMSE), mean absolute error (MAE) and R score for performance evaluation. Also cross-validation concept was introduced to avoid over-fitting and ensure generalization to unseen data.

We finalized the tools and technology that is required to perform the project. Python programming language was chosen due to the availability of huge number of libraries that would help in the analysis and implementation of ML models. Some of the useful libraries include NumPy, Pandas, Scikit-learn, Matplotlib and Seaborn for visualization, data processing and ML implementation respectively. The development environment used was Jupiter notebook for interactive implementation and analysis.

Summary of Phase-I activities:

1. Understanding of the problem domain and constraints.
2. Acquisition and analysis of a multivariate sensor data set.
3. Application of data pre-processing techniques to clean the data.
4. Feature selection and analysis.
5. Initial model development using regression and polynomial regression to establish baseline performance.
6. Definition of evaluation metrics and validation procedures.
7. Selection of tools and technologies.

All these results achieved in Phase-I laid a strong foundation to approach Phase-II of the project. In this phase we will develop a robust system and analyze more deeply for potential practical applications.

3.2 Refinement of Problem and Objectives based on Feedback

The project went through a refining process post Phase-I based on feedback received from guides, domain experts and initial experimental results. This step was crucial for improving clarity, focus and effectiveness of the problem statement and objectives. Feedback received from experts helped identify weaknesses in the original approach and improvise overall quality and applicability of the proposed system to predict concentration of ethylene (CH) gas.

The problem in Phase-I was defined broadly in terms of gas detection using machine learning techniques. However, on receiving feedback, it was understood that the problem statement had to be made application oriented. The problem statement was narrowed down to predicting ethylene concentration using multivariate sensor readings as opposed to simple detection of gas. The scope was thus changed from a generalized gas analyzer to an ethylene prediction system that can find application in various areas like agriculture (for ripening control), industry (leakage detection) and environmental monitoring.

In Phase-I, multiple machine learning models were identified with no concrete reasoning of their selection. It was then realized through feedback that polynomial regression seemed the most appropriate choice considering the potential for non-linear relations between the sensors readings and gas concentration while also keeping the problem of model implementation and interpretation simple and understandable. A simple yet capable model thus emerged to be polynomial regression rather than employing a variety of complex models.

Objectives of the project was made more precise, measurable and applicable on receiving feedback. Earlier objectives were too general with unclear significance and thus based on experts feedback the objectives were revised and more specific aims including data pre-processing, feature engineering, modeling and its evaluation and real life applicability were defined to address the problem in an organized manner. More emphasis was laid on data quality through pre-processing since it is a critical parameter in affecting model efficiency.

Also, it was brought to notice through feedback that real life scenarios are more challenging and hence more emphasis was laid on dealing with real life issues like noisy sensor data and cross sensitivity of sensors. Thus more and more sophisticated pre-processing techniques were employed so that the model can perform efficiently even in the presence of noise in data.

More rigorous validation technique was identified as K-fold cross validation since only basic validation would not guarantee robustness in the predictions. Evaluation parameters were also standardized. MSE, RMSE, MAE and R2 score were considered to be appropriate metrics for evaluation.

Another important step that was introduced as a result of feedback was that a practically implementable solution should not be very expensive in cost and time. Hence the scope of the project was made realistic with usage of low cost general purpose sensors instead of cost-effective specialized equipment and with the aim of integrating into real world IOT platforms for real time monitoring.

Structure and organization of the report was improved as per feedback received to make the report more readable and understandable. The process of presenting methodology in the report was structured and explained more clearly.

Summary of refinement was based on following criteria:

1. More specific problem statement, namely the prediction of ethylene gas concentration.
2. Primarily, use of polynomial regression to fit the relationship between sensors readings and the concentration.
3. Revised and specific objectives to direct the work more clearly.
4. Enhancement in pre-processing techniques to account for real life issues like noise, cross sensitivity.
5. Increased validity with the use of cross validation.
6. Focus on cost effectiveness and integration in real life scenarios.
7. Enhancing the applicability.

3.3 Selection of Approach / Methodology

The most significant step is to determine the right type of approach and methodology that should be used in designing an effective system for predicting the concentration of CH gas. Considering the outcomes and results that were extracted through the literature review, Phase-I analysis and feedback enhancement, it was concluded that a regression approach-based machine learning model is the most feasible method for the above described task because it supports a wide variety of multivariate sensor inputs, nonlinear relationship capturing capability, accurate predictions, and its computationally efficient to perform a practical application.

Different types of approaches were initially considered; these included analytical approach, sensor-based threshold approach and advanced machine learning models approach. However, it was observed that analytical models like gas chromatography and spectroscopy possess higher accuracy compared to the other but at higher cost and they are complex to use and not real-time based; traditional threshold sensors which simply trigger an alert but doesn't predict the data so it was considered not to be the right model to achieve the objective of the system but the data-driven machine learning based models which can be able to analyze the sensors data and predict the CH gas concentration are preferable to be used in designing the proposed system.

Among the different machine learning models which were reviewed in the literature including linear regression, polynomial regression, Support Vector Regression (SVR), Random Forest and Artificial Neural Networks (ANN), Linear Regression is the simplest form of model, but since it assumed linear relationship between the dependent variable and input independent variables, it can’t handle the nonlinear relationships presented in sensor based models. Most complex models like SVR, Random Forest and ANN can perform well in accuracy but at a considerable cost; they requires larger computational resources, are time-consuming in their training process, and are difficult to interpret in the output. The fact about this makes these models less practical for the proposed system since an appropriate algorithm for predicting the CH gas concentration has to be computationally less demanding so that it could be implemented in any available computer and even in a microcontroller or any low-end processors to perform prediction and take necessary action in time.

Thus, the polynomial regression approach was adopted for the design of this system. This method is the best choice because it can handle nonlinear relationships quite well and is still simpler to compute, train, and interpret. It is the expansion of linear regression where additional terms are added to the regression model which are the higher powers of independent variables, hence, it would be highly suitable for the current system that deals with sensors.

The methodology comprises several stages in which the first one is to acquire multiple data of sensor readings that will be fed to the machine learning model for prediction. The data is then preprocessed for removing missing values, outliers, noisy sensors' data, and for feature scaling (normalizing sensors’ data for better performance of the machine learning model). After that the process continues with feature engineering and selection to select the features that are significant in predicting the CH concentration using correlation. In order to increase the ability of polynomial regression model in capturing higher order interactions in the data, it's good practice to add polynomial features to the model.

Subsequently, the selected polynomial regression model is trained with the preprocessed and feature-engineered data using the different levels of polynomial degrees. Careful choice of the model complexity using polynomial degree to overcome bias and variance trade-off is one important step in designing an efficient system. Model is validated using K-fold cross-validation for it to be able to generalize over the unseen data. Performance metrics such as Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE) and R score are evaluated.

After model validation, performance is evaluated and analyzed by visualizing actual vs. Predicted values, distribution of errors etc. And at the end cost effectiveness of the system and its scalability is assessed and considered to be cost-effective due to its simplicity in utilizing ordinary sensors and general machine learning approach and highly scalable to be integrated with IoT technology.

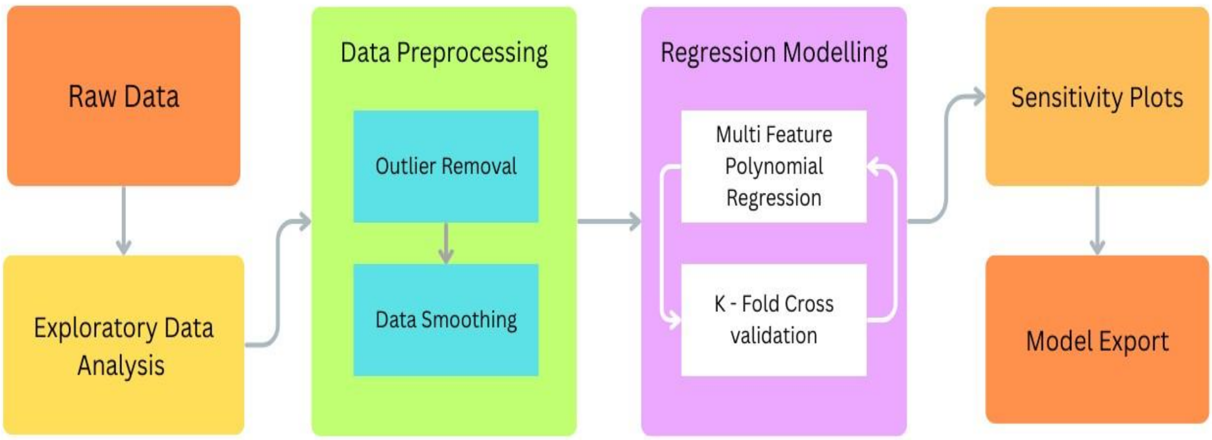

Fig 3.1: Block Diagram of the Proposed Ethylene Gas Prediction Methodology

3.4 Technical Approach and Key Elements

The technical approach taken in this project is based on the design of a machine learning-driven pipeline that predicts the concentration of ethylene (CH) gas by utilizing multivariate sensor readings. The methodology has a structured, modular design with each part-starting from the data gathering to the model deployment-working towards achieving the overall goals of the system. It also incorporates data preprocessing, feature engineering, regression modeling, validation, and result interpretation for reliable prediction. The general outline of the technical workflow can be thought of as data acquisition, exploration of the data, preprocessing, modeling, validation, and result production. The overall pipeline is designed to address the real-world complications associated with noisy sensor data, inconsistent sensor responses, and non-linear relationships that can exist between input parameters and the predicted outcome.

Data Acquisition and Exploratory Data Analysis (EDA)

The primary phase of the technical strategy involves the procurement of multivariate data which comprises sensor values taken by various gas sensors. The output of each sensor is what directly contributes to the final prediction of the ethylene concentration. EDA is conducted on the gathered data in order to establish an understanding of its organization, pattern, and other statistical elements. Diagrams such as histograms, scatter plots, and correlation matrices help to uncover existing links between variables and to spot any outliers or anomalies so that informed choices about preprocessing and model selection can be made.

Data Preprocessing

1. A very vital phase of this system is data preprocessing as sensors' measurements can vary depending on noise, outlier factors, and even environmental influences. Some of the key processes include:
2. Eliminating outliers (outliers are removed from data points using statistical techniques such as the Z-score or the Interquartile Range, IQR)
3. Smoothing data (sensor output noise can be smoothed using methods such as moving average or Savitzky-Golay filtering)
4. Scaling data (normalization or standardization is used so that input features are all at the same magnitude)
5. These procedures help to produce data that is clean, consistent and usable in machine learning systems.

Feature Engineering and Transformation

Feature engineering is utilized following data preprocessing. Correlation analysis is first applied so as to determine which features best relate to the ethylene concentration; redundant or non-significant features are later removed. Polynomial feature expansion is also performed which converts the input values into more complex features. This step helps capture non-linear relationships between sensor readings and also effectively enhances the number of features available to the machine learning model, ultimately allowing it to learn from more complicated relationships.

Regression Modeling

The basis of this system is the construction of a multi-feature polynomial regression model. This is then used with the cleaned, preprocessed data to discover the direct relation between inputs from the sensors and the concentration of ethylene. The polynomial regression model was chosen because of its suitability for fitting non-linear relationships while remaining relatively simple and easily interpretable.

Validation of Model and Metrics of Assessment

To determine the reliability and generality of the built model, a process of K-fold cross-validation is implemented where the training set is divided into K segments and tested across various configurations so as to minimize chances of overfitting. Model effectiveness and accuracy is determined by common regression metrics such as MSE, RMSE, MAE, and R-score.

Sensitivity Analysis and Visualization

Sensitivity plots and visualization diagrams also constitute an important feature of this technical design and these are employed in observing the effect of inputs on the system output. They also identify the significance of various sensor types in the prediction. Diagrams are generated as graphs that visualize actual versus predicted outputs and error distributions.

Export of Model and Deployment

The final step in the proposed technique is to save the learned model so it may be used later for predicting the concentration in real time in any application developed. While a full deployment is not in the scope of this paper, the designed model supports Internet of Things (IoT) platforms and so it can be implemented as an IoT solution.

Key features of the proposed system:

1. Multivariate sensors to enhance the model's accuracy
2. Sensors to ensure robust preprocessing and handling of outliers
3. Polynomial feature expansion for fitting non-linear characteristics
4. Multi-feature polynomial regression model
5. Cross-validation via K-fold to guarantee model reliability
6. Standard performance metrics for thorough assessment

7. Visualization graphs along with sensitivity analysis for model analysis
8. A versatile, extensible architecture for future implementation

All in all, the proposed technical approach is one which efficiently integrates both data-centric algorithms and more pragmatic considerations to achieve the overall aim of creating an accurate prediction system. The model comprises data preprocessing, feature engineering, regression modeling, and validation and provides a strong framework upon which to meet the targets set for this project.

3.5 Supporting Analysis / Design Considerations

In designing and implementing an ethylene (CH) gas concentration prediction system, it is necessary to take various technical and practical considerations into account. Such supporting analysis and design considerations are imperative for the developed system to not only be accurate, but reliable, scalable and applicable to real-world situations. Here, the design considerations guiding the adopted approach for the system design are outlined:

One of the prime considerations of this project is the nature of input data, where data is derived from multivariate sensor readings of gas sensors, which is by nature nonlinear, non-stationary, prone to noise and influenced by the surrounding environment. Hence, a detailed analysis of the dataset is performed prior to application of any machine learning techniques. This comprises statistical analysis using metrics like average value, variance, standard deviation, and data visualization tools like histogram plots and scatter plots to identify potential patterns and anomalies in the data, leading to a proper data preprocessing and modeling strategy.

Data quality is a major factor that directly affects the model performance and output. Sensor readings could possibly contain outliers, missing values and errors due to sensor drifting and calibration, as well as influences from external factors. To combat this, robust preprocessing techniques such as outliers detection via statistical methods, noise reduction using filters, and data normalization have been incorporated into the system to ensure uniformity of features' scale and avoid unnecessary influence of redundant information or noise, ensuring reasonable quality of data after pre-processing, validation by means of comparative analysis before and after preprocessing stages.

Feature selection and dimensionality reduction is another important factor to be taken into consideration as in this project we had more than one sensor reading as inputs, hence not all the features would be important for prediction. Therefore, it is important to perform a correlation analysis and choose features that have a high degree of influence on prediction. Features that were poorly correlated or that had an overlap could be discarded to simplify the model and reduce the computational complexity.

Choice of the machine learning model is another crucial design decision to be made. The chosen machine learning model needs to be capable of detecting the nonlinear relationship between inputs and the output. Following an analysis that was conducted earlier in the stages of the project, polynomial regression was deemed suitable for detection of nonlinear relationships and it offers a model that is fairly simple and interpretable compared to other complicated models. The complexity of polynomial regression in terms of the degree is another critical decision and one needs to select an appropriate polynomial degree that fits the data well without leading to underfitting or overfitting.

Validation of the model is equally as important as any other design consideration in this project. The model must not only be capable of making good predictions from the training data but also from new data, this is achieved by using the K-fold cross validation technique to determine the best fit for the model. Different performance measures of prediction, Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE) and R score are all used to get the accuracy of the developed system to predict.

Computational efficiency is another crucial factor in developing a model for practical implementation since the application requires a rapid prediction of ethylene concentration. The polynomial regression offers a good balance between the speed and efficiency of calculation. A good preprocessing strategy and choice of appropriate algorithm that will reduce computation is vital for real-time applications.

Scalability and Cost-Effectiveness is also another critical consideration to be taken into the design of this system as a well-designed system should ideally allow the use of a variety of MOS sensors available in the market, which are fairly cheap and readily available. This also reduces the over-all cost of the system and in general is cost effective for larger deployments.

Robustness to Real-World Conditions, temperature and humidity are some of the significant environmental factors that cause errors in the sensor readings which have been taken into consideration to design a robust system to counteract effects and provide accurate measurements regardless of external changes. Sensitivity analysis for various inputs is performed and the results are recorded to provide insight to the system.

Integration capability with Modern Technologies such as IOT platform so that readings can be transmitted for real time monitoring for applications such as smart agriculture, modern industry and home automation are also explored for the overall enhancement of system design and practicality. Model interpretation can also be important as some applications would require explanation of why a certain prediction was made, something which models like deep neural networks or other black box models cannot offer with sufficient ease of interpretation compared to models like the polynomial regression.

In conclusion, the overall system design and architecture is structured such that the system offers accuracy, is efficient and is flexible enough to be implemented in varied scenarios and this system provides the foundation for development of a well-established ethylene gas concentration predicting system.

3.6 Tools / Techniques Used

For successful realization of the ethylene(CH) gas concentration prediction system, a mixture of software tools, programming languages, and machine learning techniques are used, selected for their efficiency, accuracy, visualization, availability, low cost, and reliability that enables the complete machine learning process ranging from data pre-processing to evaluation. The basic software tool being used in this project is a well-known Python programming language. Being popular in data science and machine learning field for its simplicity, flexibility, and a broad library system, it facilitates straightforward implementation of complicated algorithms and good support for data analysis and visualization. Easy to write and highly readable, it allows for better modularity and scalability. For the purpose of data manipulations and analysis, the library Pandas has been widely employed.

This library has strong data structure, DataFrames, which are highly appropriate for structured sensor data. It efficiently helps in the import, cleaning, transformation and analysis of data. Missing data imputation, filtering, statistical calculations are all done through Pandas, making it an imperative component during the pre-processing of data. The library NumPy has been used for performing numerical calculations and working with multi-dimensional arrays. Mathematical operations, linear algebra computations and array calculations are conveniently dealt with by the library NumPy, contributing effectively in the feature transformation phase, especially in the polynomial feature expansion and normalization. Libraries like Matplotlib and Seaborn are utilized for data visualization purposes. Matplotlib offers straightforward ways to create simple but essential plots, such as line, scatter, histograms to get insight of the data and distributions of data and variable correlation.

Built on top of Matplotlib, Seaborn offers better high-level functionalities such as heat maps and pair plots which are useful in exploratory data analysis, correlation analysis and performance evaluation by visualization of plots of actual vs prediction. The core machine learning algorithm implementation has been done using a very useful library known as Scikit-learn (sklearn), offering all the functionalities starting from data pre-processing up to evaluation of model. The polynomial regression models have been developed through linear models associated with polynomial feature transformation using Scikit-learn which has provided efficient functionalities like data splitting, K-fold cross-validation, and performance measures like MSE, MAE and R score.

Jupyter notebook is used for interactive writing and executing codes allowing easy interpretation of analysis through integration of codes and visualization. Various data pre-processing and machine learning techniques used include exploratory data analysis (EDA), outlier detection and removal, noise reduction techniques, feature engineering, polynomial feature expansion, correlation analysis. Other techniques include polynomial regression models, K-fold cross-validation for efficient generalization and validation of model. Various evaluation parameters such as MSE, RMSE, MAE and R-score are also used to estimate the performance of the prediction models. Overall the integration of these tools and techniques helps to successfully build the ethylene prediction system.

3.7 Justification of Selected Approach

Choosing the right method is essential for the successful design of any machine learning system. For predicting ethylene (CH) concentration in this project, we chose to use a polynomial regression approach with a well-defined data preprocessing pipeline. This decision was motivated by a number of factors ranging from the data nature to system requirements and practical constraints:

The characteristics of the sensors are one of the key reasons for our choice. Data coming from the sensors is generally non-linear due to factors such as ambient temperature, humidity and the cross-sensitivity to multiple gases (more noticeable in case of MOS sensors). The simple linear model is inadequate for capture such non-linear relation leading to undesirable prediction accuracy. Polynomial regression allows modeling such non-linear trends efficiently through the addition of polynomial terms.

Model performance and complexity balance were also significant considerations during our selection process. Models such as SVR, Random Forest, and Artificial Neural Network can achieve good predictive accuracy; however, they often have a greater number of parameters and require complex setup and more computational power compared to the chosen method. Polynomial regression offers a simpler and more intuitive way to get acceptable results with less computational resources.

Model interpretability is another relevant consideration. In most real-world applications, it is important to understand the relationship between input features and prediction; hence, we selected a method whose inner working can be easily interpreted. Polynomial regression provides a clear functional mapping between features and prediction allowing for the interpretation of contribution of each feature. Complex models are often black boxes and their internal decision-making processes are hard to unravel.

The computation efficiency of the polynomial regression model is also vital as we target a system that could be utilized by an embedded system and possibly in real-time applications. Polynomial regression requires much less computation power as compared to most complex models enabling it to be used in devices with limited computational resources.

A data preprocessing pipeline was designed as well to compensate the quality issues. As most sensors have faulty readings, outliers, and noises, data preprocessing is a crucial part that is needed to get good result for the trained model.

Well-generalized solution on an unseen data was also a prerequisite. In order to achieve this, we used cross-validation (K-fold) to tune the model without overfitting, and to evaluate the performance of our system using parameters such as MSE, RMSE, MAE, R score.

Scalability and cost-effectiveness of the chosen method are another advantage. Polynomial regression can easily be used in scalable solutions without much resource requirement as compared to an ANN. The designed system utilizes easily available sensors rather than advanced expensive devices, which helps to reduce overall cost.

Additionally, we also considered expandability in the future. Polynomial regression serves as a starting point. If it is found that its performance is not sufficient for future tasks, we will always have the option to use more advanced methods in future.

Finally, the choice is well aligned with the goals and objectives of the project, and directly address challenges mentioned in previous works as regards non-linearity, efficiency, accuracy and feasibility.

Overall, the choice for a polynomial regression and data preprocessing pipeline is a wise one because of its ability to effectively manage non-linearity of sensor data, providing an optimum balance between model complexity, prediction accuracy, computation efficiency and interpretability that are desirable in a practical system.

CHAPTER 4

DEVELOPMENT AND IMPLEMENTATION

4.1 Implementation Plan and Phase-II Progress

Phase-II focuses on building a fully functional machine learning system that can predict ethylene (CH) gas concentration, from the theoretical and design phases. In this stage the entire design is tested, fine-tuned and evaluated to reach the desired performance level, following a structured implementation plan to achieve project goals in a systematic manner. The implementation plan follows a pipeline where each of the stages of a typical machine learning process is implemented.

Each of the stages in the pipeline is discussed in the following sections in order: data loading, data preprocessing, feature engineering, model development and training, validation and evaluation, plotting and results visualization and finally model tuning and export.

The first stage in the implementation plan involves the preparation and loading of the data. The multivariate sensor reading dataset was loaded using the Pandas library, and it was checked for validity, structure, and any missing values to ensure data quality and integrity prior to data preprocessing and further processing.

Data preprocessing follows the loading of data. During this stage the preprocessing techniques identified during the conceptual phase I were detailed in implementation, so outliers are removed by using the Interquartile Range (IQR) methodology. Sensor data smoothing techniques were utilized to minimize sensor noise. Standardization was done to ensure that the scale of different features do not affect the prediction performance of the machine learning algorithm by normalizing or standardizing the values of each of the sensor readings to fit in a smaller range.

Next, data manipulation, feature engineering, transformation and selection is done by computing correlation coefficients to identify important parameters influencing ethylene prediction, eliminating redundant parameters, applying polynomial feature expansion, and creating an improved dataset from the already processed data for training the machine learning algorithm and enhance its predictive capability.

The stage involving machine learning algorithm development and training is central to Phase-II. A multi-feature polynomial regression model was built and trained on the processed data using the machine learning toolkits available in Python Scikit-learn. The data was divided into a training set and testing set. The polynomial degree of the model was adjusted to attain best underfit and overfitting balance.

Model validation and evaluation was the next stage, to ensure the performance of the model trained was generalized across different samples and not biased towards the test set in the data. K-Fold cross-validation was performed along with computation of Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), R score as performance matrices of the model.

Results visualization and analysis was the following stage in the implementation plan. Plots were generated for actual values versus predicted values, prediction errors and their distribution, and heat maps showing the correlation among variables. The plots help in understanding the performance, identifying patterns in prediction error, and relationship between variables in a clear visual manner.

Finally, model tuning and exporting for future usage completes the pipeline for phase-II. The model is fine-tuned for better performance and saved as a file that can be loaded easily for deployment at later stages. Full deployment for the model has not been done yet; however, the setup is ready for integration with various real time applications and IoT devices.

Progress during Phase II

A successful implementation plan has been completed for Phase-II of the project, achieving all of its objectives. The data processing pipeline has been efficiently implemented, producing a clean and well-organized data set. The data had successfully undergone through the process of feature selection and engineering using correlation coefficients and polynomial expansion. The developed machine learning model for ethylene prediction performed well with satisfactory accuracy of the predicted values and its effectiveness can be explained by the visualization performed on the same. Furthermore, model optimization for computational speed was also achieved during Phase-II, ensuring its suitability for real-time applications.

In general Phase-II has been a very successful phase as the conceptual designs of the previous stages have been successfully converted to a fully functional working machine learning system, all tasks for Phase-II were successfully completed and has helped in laying the foundation of the future stages of the project.

4.2 Development / Execution of Major Components

To develop and implement the major components in this project, each step of the machine learning pipeline has to be followed with a methodical approach to create a proficient system to predict ethylene (CH) concentration with multivariate sensor readings. In each step, a specific process is planned, executed and implemented with consideration for data properties, model needs and application relevance:

The first major component to begin with is Data Acquisition and Loading. The dataset containing sensor readings and ethylene concentration values is then loaded using relevant Python libraries like Pandas in the working environment. Each column of the data then corresponds to the respective input and output variable, and the data loaded is then used for further manipulation. Checks are conducted to confirm data types and also to look for missing values.

The second major component involves Exploratory Data Analysis (EDA). In this step, the imported data is inspected and a number of analyses and visualizations (such as histograms, scatter plot, correlation heatmaps) are performed to get a better idea of how different sensor readings correlate with ethylene concentration. Statistical measures such as mean, median and variance are computed to find out different range distributions.

The third major component is Data Preprocessing. In this stage, some or all of the following data cleaning processes are undertaken:

Handling missing values: It is very crucial to deal with the null or missing values in the data. Strategies such as removing and imputing them were explored and tested with different combinations to achieve an optimal balance.

Outlier detection: Using statistical methods such as IQR (Inter Quartile Range), the presence of extreme values are detected and removed to prevent their negative impact on the accuracy and stability of the model.

Noise reduction: Since sensor data often has some fluctuations due to different environmental factors, a moving average technique was used to stabilize the input signal and reduce the impact of noise.

Code Snippet for Correlation Analysis

```
import pandas as pd

import matplotlib.pyplot as mp

import seaborn as sb

df = pd.read_csv("Final_Data.csv")


mp.figure(figsize=(20,12))

dataplot = sb.heatmap(df.corr(), cmap="YlGnBu", annot=True)


mp.show()
```

Feature scaling: Since it is important for the input features to have a similar scale, the data was standardized and normalized. The standard scaler was chosen because it maintains the relative relationship between input features and ensures the gradient descent converges faster.

The fourth major component is Feature Engineering and Transformation. After obtaining a clean dataset, relevant input features needed to be identified. For example, the correlation heat map obtained during EDA helped to determine how correlated each input sensor feature is with the output and also between each input features. Redundant and weakly correlated features with the target feature can be removed to reduce the complexity and improve the performance of the model. In this project, the input features were then transformed by using a polynomial function, for example, input1 raised to the power of2 and the corresponding output feature is obtained by using these higher degree terms to approximate the relation. This helps to increase the complexity of the model such as the non linear relationship between features and the target variable.

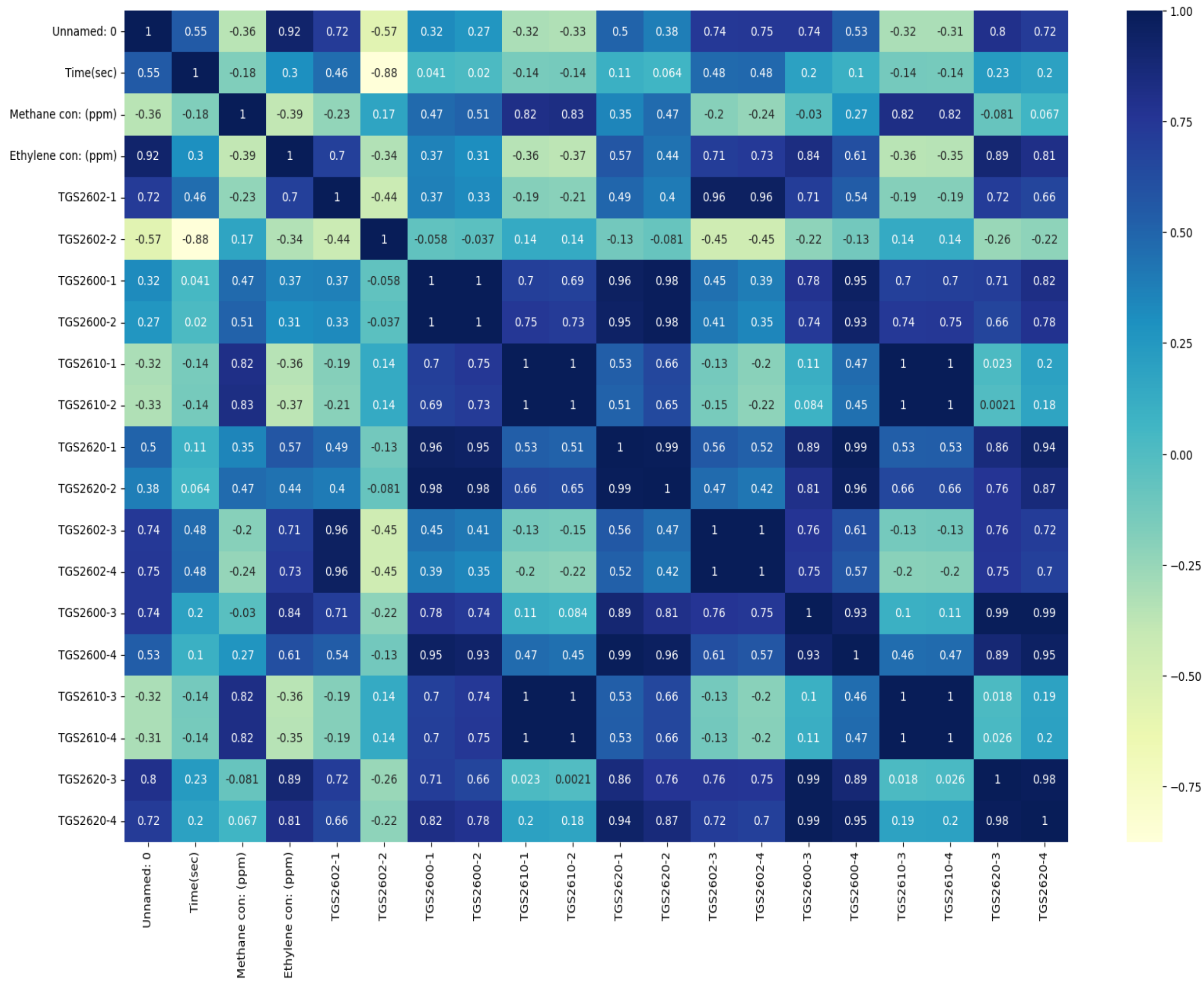

Figure 4.1: Correlation Heatmap of Sensor Features

The mathematical function representing the output based on the input is given by:

$$S_i(t) = f_i\left(C_{C_2H_4}(t)\right) + \varepsilon_i(t)$$

Where :

$S_i(t)$ = Sensor response

f_i = Sensor – specific response function

$C_{C_2H_4}(t)$ = Ethylene concentration

$\varepsilon_i(t)$ = Noise or environment variations

The fifth major component involved model development and training. A multi-feature polynomial regression model was implemented using Scikit-learn. The dataset was then split into a training set (to train the model) and testing set (to test the model). A number of polynomials of different degrees were experimented with to determine the best fit and the corresponding models were trained using the training data. This ensured that an appropriate balance between over fitting and under fitting the data was attained.

The sixth major component included model validation and evaluation. To obtain a reliable estimate of the performance, K-Fold cross validation was used on the dataset. For all the trained models, the predicted values were compared with the actual values to evaluate model performance based on performance metrics such as MSE, RMSE, MAE and R2 score. All the models were compared based on these metrics.

The seventh major component consisted of result visualization and interpretation. The predicted values of ethylene concentration was plotted against the actual values obtained by using a scatter plot. Various statistical graphs were also used to gain more understanding of the models and their respective performance such as distribution of error etc. Correlation heatmaps was also utilized again to confirm if there is any collinearity between input features which could potentially pose a problem in modeling.

The last major component of this project involved model optimization and storage. After a number of experiments, an optimized model was achieved that provided satisfactory results and could perform better predictions compared to other tested models. The best model was saved using 'pickle' so that it could be easily utilized at a later stage and used with new incoming sensor readings to make real time prediction of ethylene concentration for example at different application platforms.

In conclusion, every major component discussed above ensures that a good, robust, scalable, and efficient system is created by taking into account all potential limitations encountered with the given data and designing a solution that can perform as per the requirement.

4.3 Integration and Overall Functioning

The last step in developing our system is the integration of all of our individual modules in order to make an efficient and functioning pipeline to predict ethylene (CH) gas concentrations. In this section, we will demonstrate the interplay between the different components-data preprocessing, feature engineering, model training, and prediction-that enable the system to produce useful and accurate outputs. Integrating all of our components into a single system allows our algorithm to run as one end-to-end solution.

The system takes input in the form of raw multivariate sensor data. This usually consists of values from multiple gas sensors and may include noise, outliers, and deviations caused by environmental factors. Once data is passed through our pipeline, it reaches the data preprocessing module first.

```
# Savitzky-Golay filter
def apply_savgol(series, window_length, polyorder):
    return savgol_filter(series, window_length, polyorder)

window_length = 20
polyorder = 3

# Apply filter to selected columns
for col in columns_to_filter:
  data_frame[col] = apply_savgol(data_frame[col], window_length, polyorder)
```

Fig. 4.2 : Savitzky-Golay Filtering Function

In the data preprocessing module, various data cleaning operations are performed in order to enhance the quality of the input data. Missing values are imputed, outliers are removed using statistical techniques, and the data is smoothed to remove any noise and irregularities in the reading. Feature scaling is then carried out to bring all of the variables to an equal scale so that every input to the model weighs equally when training. It is extremely important to have clean and scaled data going into our model.

After our data has been processed and cleaned in the data preprocessing module, it is passed onto the feature engineering and transformation module. In this module we select our most informative features by use of correlation analysis, and apply polynomial feature transformation to expand the number of input features with nonlinear relations to one another. Once the dataset has been appropriately transformed, it is ready to be passed onto the model training and prediction stage.

The next step is our model training and prediction stage, where we apply our polynomial regression model. We use the transformed features as inputs to train the model, and learn the relationship between the input features and the desired output value (ethylene gas concentration). Once the model is trained, it can be used to make predictions on new or un-seen data.

The model validation and evaluation stage comes after the prediction has been made by the model. We use a variety of methods, including K-fold cross-validation, to evaluate how well the model has generalized to new data. We then compute statistics such as the MSE, RMSE, MAE, and R-score to quantify the accuracy of our model.

One very important consideration for the overall functioning is how data is passed between modules. Every module in the pipeline must take the previous module's output as input and pass on the produced output as the input to the next stage. For example, the pre-processed data goes into the feature engineering stage, and the transformed features then go into the model training and prediction stage.

We also incorporate visualization and result interpretation into the system. The graphs will be presented in ways that allow for analysis of our results including correlation heatmaps, actual vs predicted graphs and graphs that depict the error distribution for an illustration of how well our model performs.

The system is implemented using the Python programming language. Each component will be connected through an integrated script or pipeline with the use of libraries such as Pandas, NumPy, and Scikit-learn.

The final output that is generated from our system is the predicted ethylene gas concentration which can be utilized for a variety of purposes, such as monitoring of storage of agricultural produce, leak detection in industries, or atmospheric monitoring. Our prediction algorithm would also deliver the predicted results quickly in near real-time situations.

We also envision our system to be scalable and flexible to work with additional sensors, increased amount of data, more advanced machine learning algorithms in the future and to also include IOT connectivity for seamless remote monitoring and data acquisition.

4.4 Challenges Encountered and Solutions

Several technical and practical challenges were identified and addressed during the development and implementation of the system that predicts ethylene (CH) gas concentrations. These issues arose at various stages of the project-ranging from the data handling and pre-processing of input features to model design and performance assessment. Solutions were developed to create a robust and accurate final system. This section highlights the main challenges encountered, along with the solutions implemented to overcome them.

Handling noisy and inconsistent sensor data: This was a common challenge as the raw gas sensor readings, especially from Metal Oxide Semiconductor (MOS) sensors, were often affected by variations in temperature, humidity, and other parameters, along with the sensor drift which leads to noise in the signals. This causes inaccuracies and fluctuations that must be stabilized to improve model performance.

Solution: Pre-processing techniques like smoothing (e.g. Moving average) and normalization were utilized to reduce noise and scaling of input features respectively. This significantly boosted the quality of the input data.

Dealing with outliers in the data: Data often contains some irregular observations or extreme values which deviate significantly from other data points. When these are included in the learning process of a regression model, they can pull the learned model in unwanted directions and result in poorer predictions, especially when they are large in magnitude.

Solution: An outlier detection method based on Interquartile Range (IQR) was implemented and extreme values were filtered from the dataset. With a cleaner dataset, better model training can be ensured.

Handling the nonlinear relationships between sensor values and ethylene concentration: Most gas sensors produce an exponential-like output relation to the gas concentration, meaning a linear model was generally unsuitable for representing these interactions. As expected, linear regression performed poorly as it could not sufficiently capture these relationships.

Solution: Polynomial regression was introduced and used to approximate nonlinear functions by introducing higher powers of features as input. With additional non-linear terms available to learn the data from, performance significantly improved.

Managing feature redundancy and high-dimensional inputs: When features are correlated or contain the same information it leads to a higher dimensional feature space with more processing required during training. If these related features are added to the model they do not contribute significantly and the performance can be degraded.

Solution: By performing Correlation Analysis the number of features was reduced and redundancy was decreased in the system to provide a simpler but highly efficient input set of features to the regression model.

Overfitting: High-degree polynomial features were utilized for the non-linear model. However, using more non-linear feature terms leads to the risk of the regression model learning the specific trends of the training data too closely rather than generalizing and performing consistently for unknown inputs.

Solution: K-fold cross validation along with a controlled level of polynomial features allowed the model's performance to be evaluated on unknown data consistently, ensuring that overfitting had been managed.

Evaluating and selecting the model: Evaluating the performance and accuracy of a model, whether it be linear or non-linear, is key in confirming that the models' outputs were satisfactory. In many instances when a number of values like these are being approximated it is more beneficial to study a range of evaluation scores as they may highlight areas where one may not be performing well enough.

Solution: A range of performance metrics such as Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE) and R score were utilized in the evaluation of model performance.

Computational efficiency of the system: In a real-world scenario it may be desired that the system operates in real-time, meaning that computational efficiency is crucial for reducing processing time and hardware requirements.

Solution: Using only a carefully optimized and pre-processed set of highly important features enabled the use of polynomial regression to effectively reduce computation compared to extremely complex, potentially nonlinear models which may require far too many operations to run efficiently.

Integration of system components: The system involves many parts working together, such as the feature selection process, preprocessing, model training and final prediction output; hence careful attention to the interactions of these systems and how they relate to one another is paramount.

Solution: Each part of the system was developed in modular form and thoroughly tested to ensure reliable and efficient functionality before they were combined to create the complete ethylene gas concentration prediction system.

Interpretation of results: Providing raw figures for what the system predicts is not easily understandable by most people; to clarify the system's performance it was important that results were explained well and illustrated as best as possible.

4.5 Observations

Throughout the development and implementation of the ethylene (CH) gas concentration prediction system, a number of key observations were noted at various stages of the project based on data analysis, model behavior, the effect of preprocessing, and overall system performance, which provide valuable insights into the operation of the system and the effectiveness of the approach taken:

During the exploratory data analysis phase, one of the key observations made was that sensor data was highly variable and noisy. Readings from individual sensors fluctuated significantly even under identical circumstances, reflecting environmental factors such as temperature and humidity, which showed that direct use of sensor data in modeling was impractical and preprocessing was essential. Another key observation was that a purely linear model could not accurately predict ethylene concentration, based on tests with linear regression. This led to the decision to employ polynomial regression.

Correlation analysis between sensors indicated that not all sensor features have a linear relationship with ethylene concentration; some showed strongly positive or negative relationships, while others were weak. This highlighted the value of feature selection, and irrelevant or redundant features were removed to improve efficiency.

Preprocessing the data, through outlier removal, noise reduction and feature scaling had a significant positive impact on the system's performance, with the resulting predictions being more accurate and stable than those generated using the raw sensor data. It demonstrated the significance of feature engineering.

During model training, it was observed that higher-degree polynomials offered better accuracy at the cost of underfitting and overfitting. After testing several polynomials, an optimal degree was determined through experimental analysis. K-fold cross-validation produced more robust and reliable performance estimates of the system when applied compared to just a train/test split of the data. A combination of MSE, RMSE and the R score all provided useful metrics for measuring the performance of the model.

The visualization of predicted against actual values showed that the model had accurate results; however slight variations were observed where values were higher. Error plots showed that most predictions had only small errors.

Compared to other potential solutions, polynomial regression offered a good compromise between system performance and computational resources required to train the system. By breaking the system down into individual modular components, development and debugging became much simpler, increasing the flexibility of the system and the speed at which subsequent stages could be refined. One of the final observations made during the implementation phase was that the developed system is suitable for deployment in several environments such as agricultural settings or for industrial safety applications due to its ability to predict ethylene concentration with reasonable accuracy.

CHAPTER 5

RESULTS, ANALYSIS AND VALIDATION

5.1 Testing/Evaluation Methodology

The approach adopted for testing and validation of the model to estimate the concentration of ethylene gas (CH) is critically important to evaluate performance, reliability, and generalization ability of the model. Machine learning based approach requires specific systematic process of testing and validating to evaluate performance of model on training and unseen data. Strategies, techniques, and performance parameters to test and validate the model built is given in this section.

The first process involved in the testing process is the splitting of the data-set into training set and testing set. The training data is used for training the polynomial regression model such that it can learn the correlation between sensor readings and concentration of the ethylene gas, whereas the unseen test data is used for testing the performance of the model developed so. This test set helps in checking the generalization capability of the model.

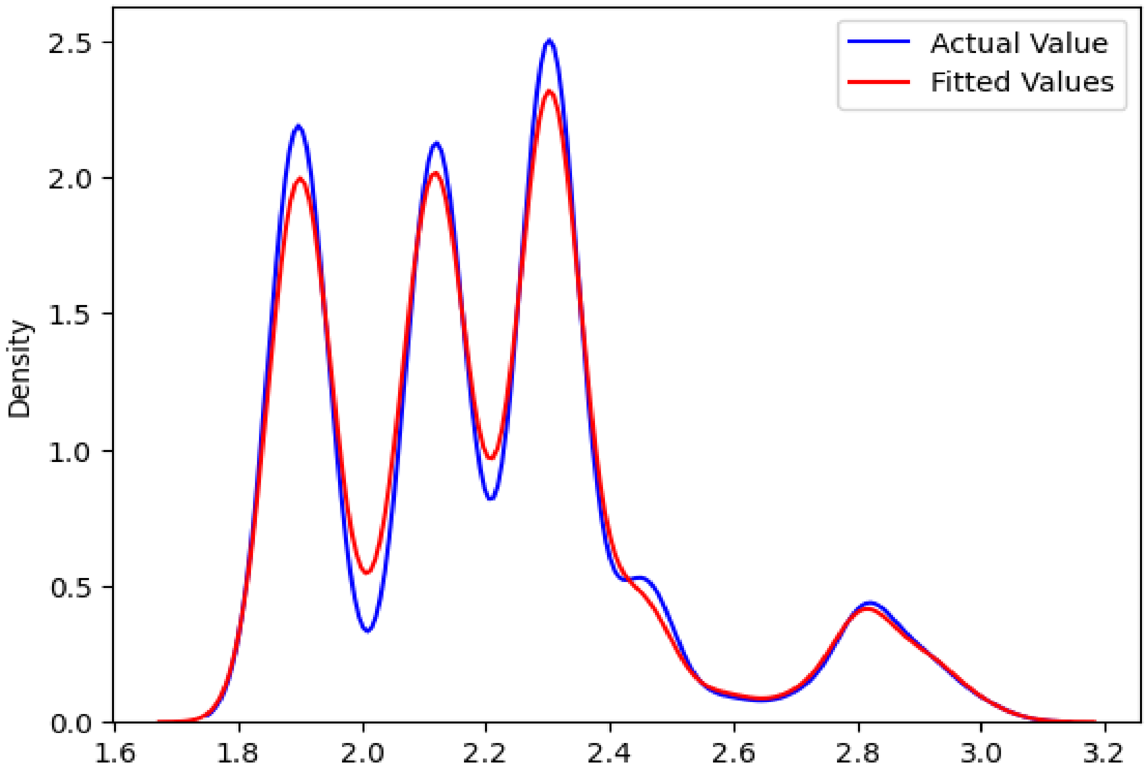

Fig.5.1 : Distribution Plot representing actual and fitted values

In order to get more robust and reliable results K-fold cross validation is used. In this methodology the data-set is split in K equal number of subsets or 'folds'. The trained model is tested on K-1 folds while it is trained on remaining one fold. This process is repeated K times and at each instance one of the fold acts as test set and average of the K results is considered as final value. It reduces the chances of model being overfit and get unreliable results.

One of the most important parameter for the testing is the evaluation parameters. The following are few parameter used for regression problem to check how close the predicted value to actual value is.

Mean Squared Error (MSE): - The average of the squares of the errors or deviations (predicted value minus the actual value). Lower value implies better performance.

Root Mean Squared Error (RMSE): - Square root of the MSE and it provides the error in the same unit as the actual value of the target. Lower value implies better performance.

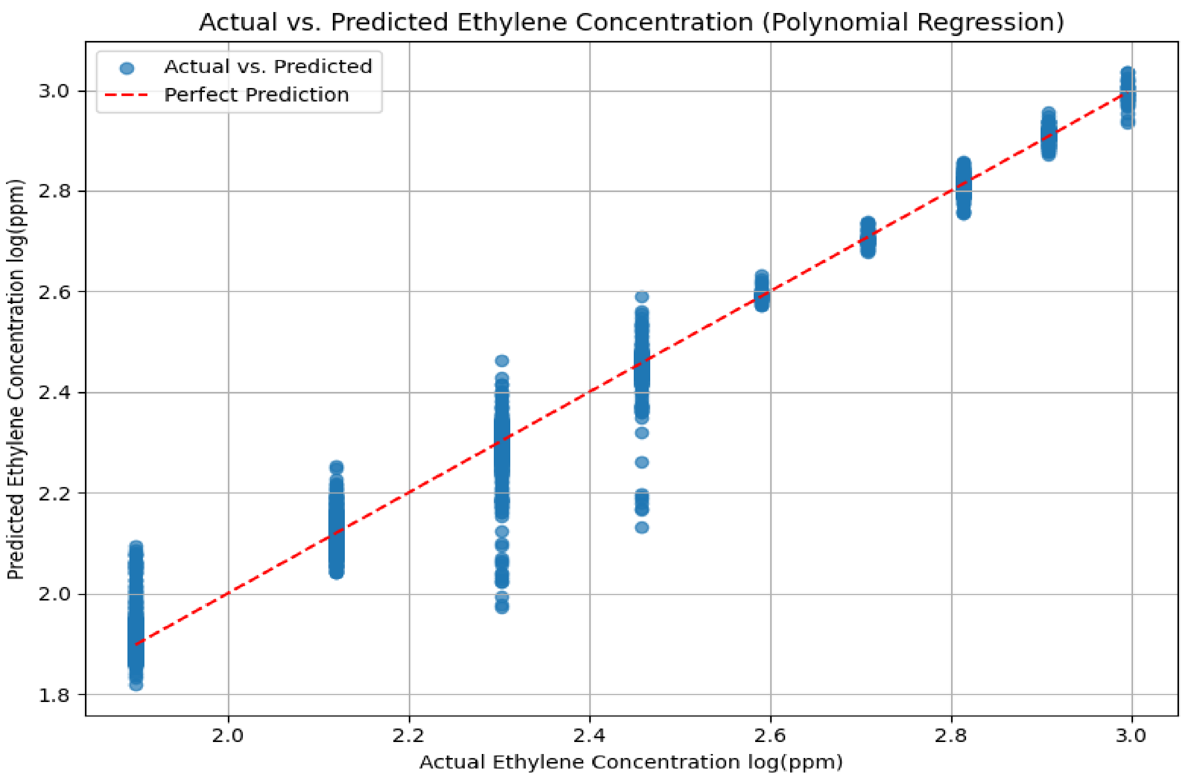

Fig. 5.2 : Actual vs Predicted Concentration

Mean Absolute Error (MAE): - Average of the absolute differences between actual and predicted value. This also provides a direct interpretability of the prediction error.

R Score: - Measures how well the model explains the variance in the data. Higher value close to 1 implies better performance of model.

The testing of model is also performed by visual inspection. Plots such as plot of actual value versus predicted value and distribution of error are generated which visually shows how good is the model in predicting the output. Comparison among models of various polynomial degrees are carried out to find out the best polynomial for the model. This helps to avoid over-fitting and under-fitting problems.

Real world scenario also includes the noisy sensor readings and effect of environmental factors. It can be check how well model performs when test data is effected by noise and other factor using same procedure by applying varying level of noise on test data. The improvement that could be achieved using the pre-processing methods employed on the data-set also been tested and validated.

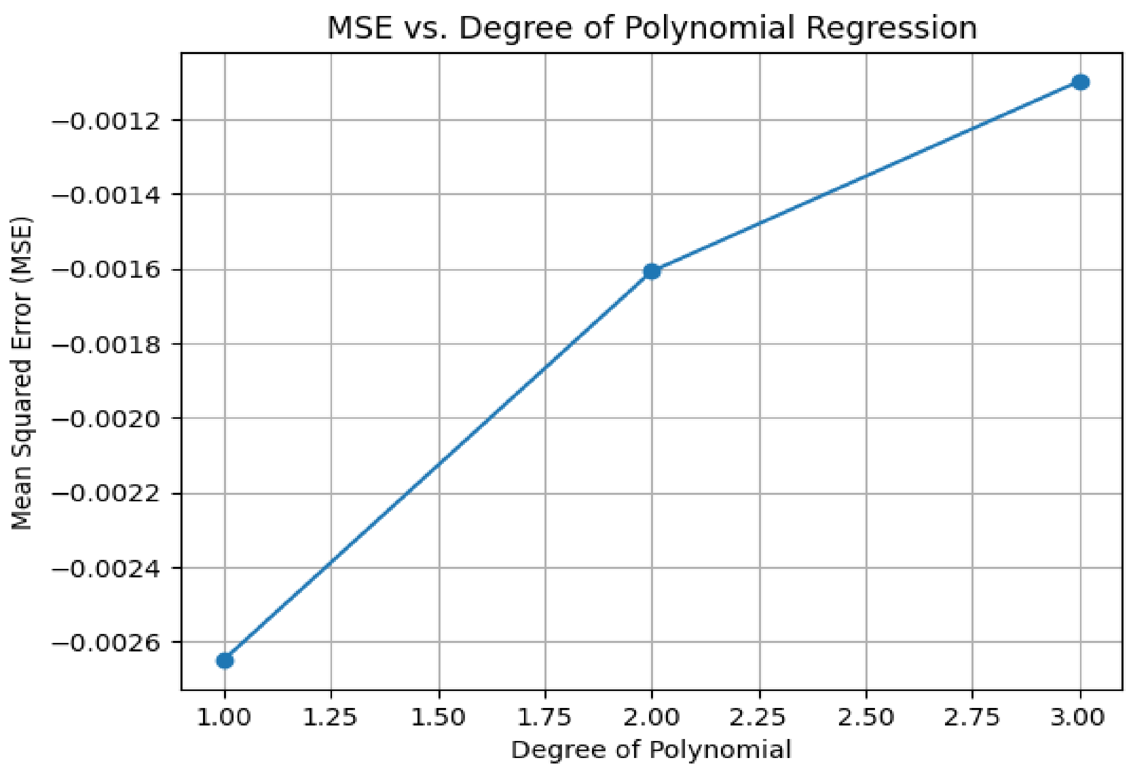

Fig. 5.3 : Relationship between Mean Square Error and degree

It is also ensured that tests were reproducible, meaning that if you performed the same test under the same conditions then the results should be consistent. This would be made consistent by setting a seed for random numbers used in data splitting.

5.2 Results Obtained

The presented results for the ethylene (CH) gas concentration prediction system indicates the capability of the proposed machine learning method. After going through data preprocessing, feature engineering, model training and testing, a polynomial regression model is applied to predict the ethylene concentration based on multiple inputs from the sensors and validated using quantitative and visual methods.

The results from the execution of the prediction system for ethylene gas concentration prove that machine learning can be utilized effectively for predicting gas concentration levels. The main result obtained after successful development of a trained polynomial regression model to learn the non-linear relation between the sensors readings and the ethylene concentration has been used to predict the ethylene concentration with previously unseen testing data and the actual data values are compared with the predicted values. The outcome of comparison show that the predicted values correspond strongly with the actual values.

```
Results :

Average Mean Square Error / Validation Loss : -0.0010990435800110224
Std. Dev. of MSE : 0.0001229416983753084
Average R-Squared : 0.9849999211915513
Average R-Squared Error : 0.01500007880844867
```

Fig 5.4 : Results figure

Metrics such as MSE, RMSE and MAE are utilized to examine how well the predicted values matches the actual values. All the metrics shown have low values which suggests that the errors in prediction are at a very minimum level and it is acceptable. Apart from that R score which defines the extent to which the model explains the variance in the given data has an R score close to 1. Hence the model explains almost all the variations within the data.

Table 5.1: Comparing accuracy of Polynomial Models by degree:

Degree	Mean Square Error (MSE)	R^2 Score	R^2 Error
1st	-0.0026487618654890666	0.9638697814210273	0.036130218578972695
2nd	-0.001609589492804099	0.978033513878405	0.02196648612159502
3rd	-0.0010990435800110224	0.9849999211915513	0.01500007880844867

Visualization with the help of actual vs predicted plot helps to study the performance of the model and also explains the closeness of the predicted and actual values. There were few deviations which may occur due to non-linear relationships.

5.3 Analysis and Interpretation of Results

This section analyzes and interprets the obtained results to give us a better understanding of how the ethylene (CH) gas concentration prediction system works. In the previous section the numerical and graph results were represented, in this section an analysis and explanation of the results obtained are made in terms of how efficient the system performs and what we can deduce from these results, which helps validate the methods used.

From the results we observe that one of the important findings is that polynomial regression is highly efficient. Mean Squared Error (MSE) and Root Mean Squared Error (RMSE) values being small, shows a difference between values predicted and those that are true that is small. This means that the model has learned efficiently the relation between sensor inputs and ethylene concentration. Mean Absolute Error (MAE) also confirms that the difference is small and uniform on average for all values predicted.

```
#Exporting the model to pkl
import joblib
with open("ethylene_polynomial.pkl","wb") as f:
    joblib.dump(pipe,f)
```

```
#Loading the model
model=joblib.load(r"ethylene_polynomial.pkl")
model
```

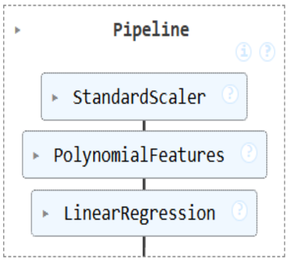

Fig. 5.5: Exporting and Importing Model for usage

The R value is almost 1. It clearly shows that the amount of variance on the data explained by the model is very high. Input features used (sensor data) are very informative for model training. The use of polynomial regression model fits well for the nature of data, which is non-linear.

The actual vs. Predicted graph shows how the values predicted fit those measured with an appropriate match in most cases. It's noticeable that the points cluster close to the y=x reference line which indicates the predictions are similar to the measured values, except in the far extremes where the predictions are less accurate compared to actual values, which is a normal phenomenon and doesn't really influence the efficiency of the model much.

Analyzing the distribution of errors clearly indicates that the errors are almost centered on the zero mark, meaning that the model doesn't systematically over or underestimate the predicted value, which in turn means that the model is not biased, its errors are balanced for the whole dataset and its performance is consistent. A large spike in errors is not present, meaning the model is robust.

It was also found out that the use of preprocessing technique has significantly affected the accuracy of the prediction results. Data preprocessing (noise elimination, removal of outliers, normalization of features) had resulted in improved prediction and decreased error values compared to results with unprocessed data. This clearly signifies that the quality of the input data has a great impact on the efficiency of any machine learning system particularly sensor based systems.

The polynomial degree selected for the model is very important as well. As it can be observed, an optimal degree helps to neither overfit nor underfit the data. An overly low polynomial degree does not manage to capture the complexity and thus results in a poor model; an overly high degree results in overfitting the model on the given data, meaning the model will work for training data but not for testing data. A good degree like the one chosen in this work ensures that the model works good on both testing and training data.

The validation results from K-fold cross-validation have also indicated that the system is consistent and does not rely solely on a specific part of the dataset, meaning that the performance is very reliable on future, unknown data as well.

In a real world scenario, this system has proved to be accurate and efficient enough for the accurate determination of ethylene concentration values. Due to the multi-feature and accurate prediction abilities of the system, it can be applied to farming for monitoring growth of fruits, agriculture etc. Moreover, it can be utilized for monitoring of air quality and even in the safety of workers where toxic gas concentrations can be very harmful. Low computational requirements will permit implementation of this system even for real time operations.

One area that can be further worked on is the accuracy at extreme values, which could be improved by adding more values at extreme points. Another aspect is the robustness to noise and sensor malfunctions. The performance depends highly on the quality of sensors, but this is true for most systems that rely on sensor data.

In conclusion, based on the accuracy of the results obtained from various metrics used, and the analysis performed on these results; it has been concluded that the system designed to determine the concentration of ethylene gas is accurate, reliable and applicable for real-time uses. The R value near one and small error values are proof of a strong and efficient model that is not overfitted and not underfitted as observed from the polynomial degree used. The robustness is also shown by the K-fold validation performed.

5.4 Validation Of Objectives

Objective validation is a process to verify if the stated goals at the start of the project were achieved or not. The main goal of the current project is to create a highly efficient and robust system to predict the CH concentration using machine learning. The project goals as mentioned in Chapter 1 guide the entire process of developing the system. Below is a section to judge if each objective listed in chapter 1 was achieved in accordance with the results from the experiments:

The first objective of the project was to find and study a multivariate dataset that includes all the sensors and theCH gas concentration value. This objective has been achieved since a dataset containing various sensor readings as input and the actual CH gas concentration value as output has been selected. Exploratory data analysis was conducted on the dataset to discover patterns and structure within the data. Through use of graphs like a correlation heatmap, interdependency of variables can be examined; this completes the objective.

Second objective of the project was data pre-processing. This objective was also achieved by cleaning the dataset of missing values, outlier, noise, etc. And applying feature scaling to bring all the variables in a single range to get higher efficiency and consistency for model training. This was also validated by low values of errors obtained after this step, hence better predictions were obtained.

The other objective was finding out relevant features and perform feature engineering. This was completed using feature selection where important sensor inputs that has highest correlation toCH gas concentration is identified. Furthermore, the interaction features were generated by means of polynomial expansion to account for non-linear relationships that exist. Feature engineering also proved effective in improving prediction as seen by better error values after inclusion of these features.

Main objective of the project was to develop the machine learning model that will be used to predict theCH concentration value using selected sensors. This objective was fulfilled by developing a polynomial regression model, since it is known to capture non-linear relationships between variables more efficiently. Result of the polynomial regression models that the predicted values closely matches with the actual values; as shown with graph and R-score values and errors associated, this objective is achieved.

Another objective was to validate the developed model. This objective was fulfilled by using k-fold cross-validation method to evaluate the accuracy of the model on different dataset; and use of various errors metric to measure the performance of the model in detail, such as MSE, RMSE, MAE and R-score. Thus validation using such multiple parameter confirmed the accuracy and robustness of the model.

The objective was to understand the result using data analysis. Both numerical and graphical analysis were done.Graphs like the actual and predicted graph and error analysis shows the model behavior and its performance; and hence the results were interpreted correctly, which indicates that this objective has been fulfilled.

The objective was to build an efficient and real-time prediction system that is capable of making efficient prediction. The polynomial regression model established ensures that prediction is fast enough, for example a prediction can be made within a few milliseconds, thus it is suitable for near real-time predictions; additionally it is easy to implement in real world applications like smart systems by integration with various IoT devices.

Finally, the last objective was to document the process of the whole project thoroughly and provide the entire work on the report. Each step from selecting a problem to analysis of result has been properly explained and documented in this report, thus completing the objective.

To sum it up, all objectives listed above were successfully achieved. The predictive model can produce highly accurate and robust prediction value for the CH gas concentration. Thus, the project achieved its aim successfully.

5.5 Comparison with Expected/Existing Outcomes

We will evaluate the system proposed in this project for its performance by comparing the predicted values with the real ones, as well as with standard methods. At the start of the project, we expected the system to provide a good estimation of ethylene concentration based on a multivariate sensor data and a trained model, without significant loss of precision, especially in low ranges, providing stable results.

As the results obtained in this project demonstrate, this system has met our expectations for its performance. The low error values of MSE, RMSE, and MAE and their very small fluctuations within different samples and the closeness of predicted values to the true ones make this system quite good and efficient.

In comparison with current gas detection systems, the proposed one has significant improvements. Most of the traditional systems only generate an alert when the concentration of gas reaches a specific threshold, making them reactive rather than proactive. The system presented here is proactive, it tries to predict the concentration before the gas itself reaches dangerous levels.

It was also compared to a linear regression method, and results showed a substantial difference for the nonlinear behavior of gas concentrations according to sensors. The proposed method based on polynomial regression was better for the behavior of data from sensors.

When compared to an array of other more complex methods such as SVR, ANN, and Random Forest algorithms for data regression, it should be said that this proposed method, in some conditions and applications can achieve the desired results, with very less complex calculations and more understandable algorithm compared to ANN or SVR. Also the training was very fast for our case. Our results achieved reasonable R values, meaning that the variance of the error, was lower, than with basic methods and that this system can be a good substitute for applications that require a rapid analysis, without high accuracy demanded.

The values obtained were analyzed according to typical error metrics for regression problems such as the Mean Squared Error (MSE), the Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE). For the different tests carried out the mean errors obtained are very low, representing the accuracy of the system, while R scores, being near to 1 demonstrate that a high percentage of the variance of the data was explained by the model. Comparing to the literature, it is considered a good result, even if some of the existing methods (like advanced neural network systems) can reach even better accuracy on larger data set. The model explained most of the variance in data and error value remained minimal.

The effect of data preprocessing was also taken into consideration, and we found that it strongly influences on overall performance. Contrary to most of the approaches presented previously, our method included intensive preprocessing stages that removed most of the errors and made data suitable for the model.

Another factor which must be taken into account is the cost and the scalability. Traditional high-precision detection techniques (as the gas chromatograph) are very expensive and cannot offer continuous and real-time monitoring in industrial processes or in the fields. The proposed system uses very simple sensors that are relatively inexpensive and allows real-time monitoring at a much lower cost, hence this system can be used on larger scales for more applications like green houses and warehouses.

Nevertheless, we must admit that with a more complex model and deeper neural networks, it is possible to achieve slightly better performance, as a lot of models do not show any loss for extreme values, and are very accurate for this case too. Despite, the proposed system has reached its goals of performance and prediction.

In conclusion, comparing with its expected objectives and other research, the proposed system for the detection of Ethylene concentration offers promising results; it is efficient and accurate, while computationally economical, therefore making it suitable for many practical applications.

CHAPTER 6

CONCLUSIONS AND FUTURE SCOPE

6.1 Conclusions

The work completed in this project clearly establishes the success of using machine learning techniques in the prediction of Ethylene(CH) gas concentrations based on a multivariate sensor setup. A principal aspect of this project was to construct an efficient, robust, accurate and intelligent prediction system that could resolve some of the limitations associated with conventional gas detection. After thorough consideration of the implementation, results and analysis that were made throughout previous chapters, it can be said with a good degree of certainty that the project has fulfilled the objectives stated and intended to address.

One of the main conclusions of this project is that a machine learning approach is an efficient way to handle the complex and non-linear interactions found in sensor readings. Unlike a system using a threshold-based detection system, the implemented model allows for a prediction of the ethylene concentration based on readings from various sensors, hence enabling a preemptive detection system to be formed. This is most important in an application such as agriculture where it is crucial to detect Ethylene levels in fruit storage so that the fruits are not over-ripened.

Another conclusion that can be made is that data pre-processing is a critical stage that greatly aids in improving the accuracy of a model. Since sensor data often is influenced by environmental factors and noise, accurate readings can not always be assumed. The pre-processing methods applied (outlier detection and removal, data cleaning, noise filtering, and scaling of values) created a more uniform set of data for the model to work with.

Also, it can be deduced that polynomial regression was an efficient algorithm for this task. This was since unlike a linear model, which is unable to capture the complex relations between the readings from the various sensors and the ethylene concentrations, it was found that an exponential model was able to achieve an acceptable level of accuracy and accuracy, as illustrated in the achieved R score and errors.

It is also important to consider that feature engineering and selection played a significant part in increasing the accuracy of the system. With an initial analysis of the correlation between the various sensors and also based on experimentation, irrelevant sensors were excluded to improve model efficiency and performance. In addition to that, using polynomial features enhanced the ability of the system to better handle the non-linear nature of the readings.

The use of k-fold cross validation proved to be important in establishing a model with minimal risk of overfitting to a single set of data. The results from each of the validation fold produced fairly low errors, confirming the stability and reliability of the model.

In terms of implementation, the use of Python and associated libraries was particularly useful. Libraries such as Pandas, NumPy, and Scikit-learn, when combined with an appropriate system design, allow for a smooth development process, a quick method of implementation and efficient analysis.

In regards to the implementation, another key conclusion is the practical use of the developed system. The system requires minimal computation and hence is very adaptable to being implemented in a real-world environment where the ethylene concentration could be monitored in a fruit storage unit, a food packaging system or an environmental monitoring system. With inexpensive sensors, such a system would also be extremely cost-efficient.

Some of the limitations were observed, mainly concerning the slightly poor prediction at values that were in the extremities of the range, and the fact that accuracy is heavily reliant on good quality sensor input. Both of these factors can lead to an increase in the overall error of the system.

Therefore, it can be concluded that the project has successfully designed and implemented a machine learning-based system for the prediction of ethylene gas concentrations using a multivariate sensor arrangement. Through the use of polynomial regression along with appropriate pre-processing and validation steps, the system is able to yield predictions with a reasonable level of accuracy and efficiency.

6.2 Limitations of the Work

Although the presented ethylene (CH) gas concentration prediction system shows impressive and promising results, there are some limitations that exist with the current work. In an effort to provide a realistic interpretation of the capabilities and constraints of the system, the following limitations were considered.

A major limitation to this project was the dependency on data. The performance of the model was directly proportional to the quality and comprehensiveness of the data used to train and test it. Since the system relied on the readings of sensors it was inevitable that the data would have errors, missing or duplicate information and that the accuracy of the model would also vary accordingly. Although efforts were made to pre-process the data as much as possible the data from sensors will inherently be susceptible to noise and errors.

The system was trained on a particular dataset so as to ensure the validity of the model performance on that dataset. As it stands this may not necessarily mean that it will have the same results in different environments or on different data-sets. There is a degree of limited generalisability.

The system showed some limitations when predicting high and low values, i.e. Extreme values were subject to slight inaccuracy. This was possibly due to lack of high or low extreme data points in the dataset and as a consequence it may have failed to train on these values.

Another limitation to this system was that polynomial regression was chosen as the primary model to be used for the system. It gave a good balance between accuracy and complexity but it is certain that in cases with greater complexity there would be a better performing model e.g. A deeper learning model but these models are much more complex and would have been computationally demanding.

The system has some limitations in terms of implementation in a real world scenario. The model in this project is capable of being near real-time as is has low computationally demand and it is the only main issue holding it back from fully real-time implementation is the lack of the connection to a data logging system e.g. A network of IoT sensors.

It was noticed that the proposed model failed to account for any environmental parameters (e.g. Temperature, humidity, pressure) that could affect the readings from the sensors and thus effect the final prediction. The model should be capable of taking these parameters into consideration.

The model did fail to take into account the possibility of sensor limitations e.g. Cross-sensitivity or drift of the MOS sensors over time that would alter the raw data being read and therefore altering the system performance.

Finally, another limitation to this project could be mentioned that is a limited choice of evaluation metrics was used, only common metrics like MSE, RMSE, MAE and R were used, however real time performance, accuracy with regard to environment and more domain specific metrics could have provided an additional performance assessment.

In conclusion it is evident that the developed system has met the requirements of its project scope. However as with all projects there are always limitations to what has been developed and in this instance those limitations include, data dependencies, reduced generalisation across different datasets, inaccurate high/low predictions, no real-time implementation, limitations with model and sensor design.

6.3. Future Scope

The developed ethylene (CH) gas concentration prediction system shows promise but has significant room for improvement and expansion. Areas that can be explored in future work include enhancing model accuracy, expanding the system functionality, and deploying the system in real-world applications. The following is a discussion of such opportunities:

Use of sophisticated models:

The polynomial regression has produced favourable outcomes. However, machine learning and deep learning models such as Random Forest, Gradient Boosting, Support Vector Regression (SVR), and Artificial Neural Networks (ANN) should be used and tested for better prediction. These models can handle non-linearities much better if the available data size is large and diverse.

Real-time data acquisition system:

The current implementation is based on offline data and requires offline calculation. A real-time data acquisition system could be designed in the future. Internet of Things (IoT) devices and sensor network would be incorporated to have real-time monitoring and prediction of ethylene gas concentration in an industrial environment.

Inclusion of additional features:

Input features can be extended to include temperature, humidity, pressure, etc. As they are important factors that affect the response of the sensor and the concentration of the gas.

Data expansion:

The system can be tested with large and diverse datasets collected under different conditions and sensor types. This would help in better training of the model so it can perform better for even extreme cases, where sensor reading at the boundaries is important to predict.

Multi-gas prediction:

The developed model can be trained to predict multiple gas concentrations such as methane, carbon monoxide, carbon dioxide etc simultaneously and this system can be extended in such domains where these gases are needed to be detected.

User friendly interface or application:

A user-friendly application interface or a web application can be developed so it can be easily used by the non-technical person too.

Deployment of the system on embedded systems or cloud platforms:

The system can be deployed on microcontrollers or edge devices for on- site prediction with low latency. Alternatively, it can be deployed on cloud-based platforms for real time data collection and processing, making the system scalable.

Automated model updating and learning:

The system could be designed in such a way that it updates itself whenever new data becomes available and its accuracy is maintained in real time application

Sensor drift and calibration:

Adaptive calibration and drift compensation mechanisms can be introduced in the system to minimize the error occurring due to drift in sensor reading over time and the collected data can be used to create accurate predictions in the long run.

Hybrid models or ensemble methods:

Combining different ML models to enhance performance and stability.

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